Department of Mechanical and Production Engineering

Aarhus University

Lecture Notes

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Contents

Lecture 1: Introduction 17. Marts 2026

1 Course Information

These are the personal notes, written by Noah R. B. Hansen for the course in Control Theory at Aarhus University taught by Xuping Zhang, PhD. as part of the BSc in Mechanical Engineering.

2 Basic concepts and introduction of control theory

Definition 1: Control System

A *control system* is a device, or a set of devices, that manages, commands, directs, or regulates the behavior (output) of other device(s)/system(s)/processes by adjusting the input. For an example of a control system refer to the bottom of Figure 2.1

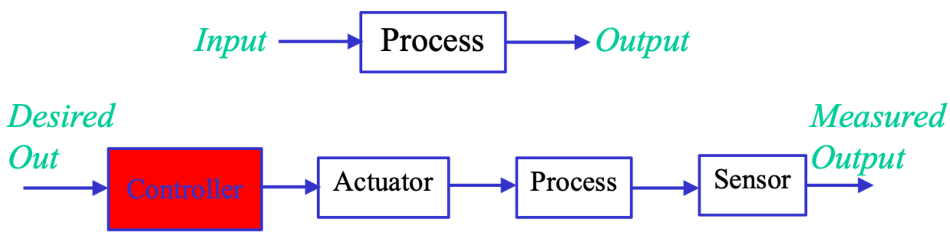


Figure 2.1: Comparison of a simple uncontrolled process and an example control system.

In Figure 2.1 two systems are shown. The top figure is an uncontrolled process that receives an input and outputs something without concern for anything other than how it is initially set up.

In the bottom diagram, the process is part of a control system. Here the desired output is fed to a controller that determines what action should be taken to match the real output to the desired one. The system is then fed to an actuator that applies the commands of the controller to the system, e.g. a motor, valve, heater, pump, brake, etc. The system is then fed to the process that is being controlled. This gives a signal to a sensor measuring the state of the process, e.g. a thermometer, encoder, tachometer, barometer, etc. The output of this sensor is then (ideally) the actual output of the system.

When talking about control systems we often differentiate between open-loop and closed-loop control systems.

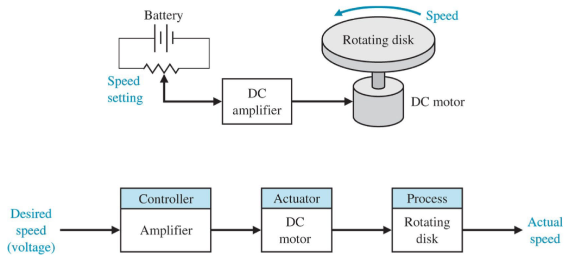
Definition 2: Open-loop Control System

An open-loop control system utilizes an actuating device to control the process directly without using feedback. An example of this is shown on Figure 2.2a.

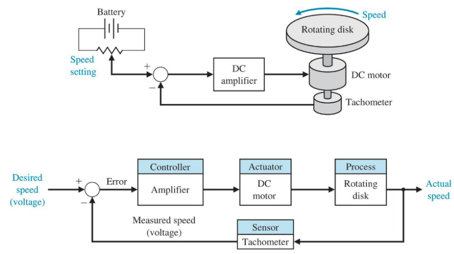
Definition 3: Closed-loop Control System

A closed-loop control system uses a measurement of the output and feedback of this signal to compare it with the desired output (a reference or command). An example closed-loop control system is depicted on Figure 2.2b

In general an open-loop control system are much simpler than as they do not require any sensors and do not introduce any stability problems on their own. However, their performance is also rather poor compared to closed-loop systems to match the desired output well.



(a) Open-loop control system.



(b) Closed-loop control system.

Figure 2.2: Comparison of an open and closed loop control system.

Closed-loop systems excel in that they do not need human adjustment and are able to respond to disturbance. These, however, may experience steady-state error or cause stability problems.

3 Description of the dynamic behavior of physical systems by Differential Equations

Many different physical systems are governed by differential equations.

Example: Spring-Mass-Damper Mechanical system

On Figure 3.1 a mass M can move vertically up or down. It is connected to a spring with stiffness k, a damper with damping/friction coefficient b, and an applied external force $r(t)$.
The acceleration of the mass is related to the net force by Newton's second law:

$$M\ddot{y} = r(t) - b\dot{y} - ky \implies M\ddot{y} + b\dot{y} + ky = r(t).$$

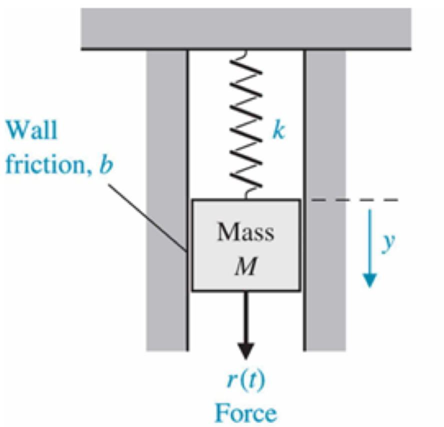


Figure 3.1: Spring-mass-damper system.

Example: Resister-inductor-capacitor Circuit Electrical System

On Figure 3.2 a parallel RLC circuit is driven by an input current source $r(t)$ and the output is the voltage $v(t)$ across the branches. Here, the resister gives a current proportional to voltage:

$$i_R = \frac{v}{R},$$

the capacitor has a current of:

$$i_C = C\frac{\mathrm{d}v}{\mathrm{d}t},$$

and the voltage-current relation of the inductor is:

$$v = L\frac{\mathrm{d}i_L}{\mathrm{d}t}.$$

If we apply Kirchoff's current law at the top node:

$$r(t) = i_R + i_L + i_C$$

and then substitute into the branch relations:

$$r(t) = \frac{v}{R} + i_L + C\frac{\mathrm{d}v}{\mathrm{d}t}.$$

Taking the time-derivative yields:

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{\mathrm{d}i_L}{\mathrm{d}t} + C\frac{\mathrm{d}^2v}{\mathrm{d}t^2} = \frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{v}{L} + C\frac{\mathrm{d}^2v}{\mathrm{d}t^2}.$$

Which can be rearranged to

$$C\frac{\mathrm{d}^2v}{\mathrm{d}t^2} + \frac{1}{R}\frac{\mathrm{d}v}{\mathrm{d}t} + \frac{1}{L}v = \frac{\mathrm{d}r(t)}{\mathrm{d}t}.$$

Note that this is analogous to the result from the mass-spring-damper system above.

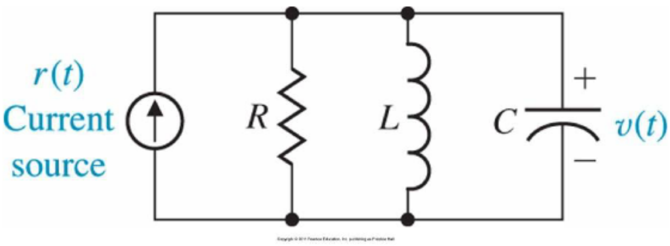


Figure 3.2: Resistor-inductor-capacitor system.

In general, many dynamic elements can be broken into inductive storage, capacitive storage, and energy dissipaters. A table is shown on Figure 3.3.

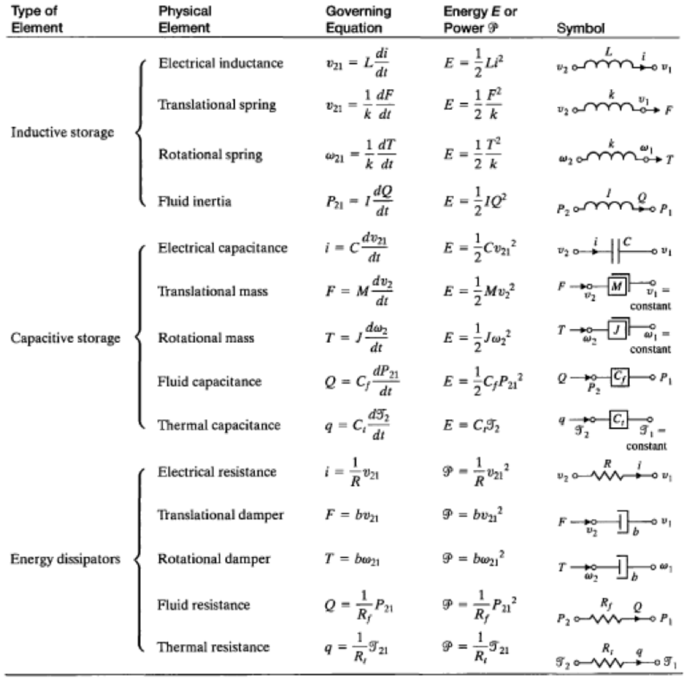
4 Linear Systems

A linear system is any system that satisfies superposition. I.e. if input u_1 gives output y_1 and input u_2 gives output y_2, then input $au_1 + b_u2$ gives output $ay_1 + by_2$ for constants a and b. This means

$$m\ddot{y} + b\dot{y} + ky = r(t)$$

is a linear equation as all y, $\dot{y}$ and $\ddot{y}$ appear linearly, but

$$m\ddot{y} + b\dot{y} + ky^3 = r(t)$$



Type of Element	Physical Element	Governing Equation	Energy E or Power $\mathcal{P}$	Symbol
Inductive storage	Electrical inductance	$v_{21} = L\frac{di}{dt}$	$E = \frac{1}{2}Li^2$	
	Translational spring	$v_{21} = \frac{1}{k}\frac{dF}{dt}$	$E = \frac{1}{2}\frac{F^2}{k}$	
	Rotational spring	$\omega_{21} = \frac{1}{k}\frac{dT}{dt}$	$E = \frac{1}{2}\frac{T^2}{k}$	
	Fluid inertia	$P_{21} = I\frac{dQ}{dt}$	$E = \frac{1}{2}IQ^2$	
Capacitive storage	Electrical capacitance	$i = C\frac{dv_{21}}{dt}$	$E = \frac{1}{2}Cv_{21}{}^2$	
	Translational mass	$F = M\frac{dv_2}{dt}$	$E = \frac{1}{2}Mv_2{}^2$	
	Rotational mass	$T = J\frac{d\omega_2}{dt}$	$E = \frac{1}{2}J\omega_2{}^2$	
	Fluid capacitance	$Q = C_f\frac{dP_{21}}{dt}$	$E = \frac{1}{2}C_fP_{21}{}^2$	
	Thermal capacitance	$q = C_t\frac{d\mathcal{T}_2}{dt}$	$E = C_t\mathcal{T}_2$	
Energy dissipators	Electrical resistance	$i = \frac{1}{R}v_{21}$	$\mathcal{P} = \frac{1}{R}v_{21}{}^2$	
	Translational damper	$F = bv_{21}$	$\mathcal{P} = bv_{21}{}^2$	
	Rotational damper	$T = b\omega_{21}$	$\mathcal{P} = b\omega_{21}{}^2$	
	Fluid resistance	$Q = \frac{1}{R_f}P_{21}$	$\mathcal{P} = \frac{1}{R_f}P_{21}{}^2$	
	Thermal resistance	$q = \frac{1}{R_t}\mathcal{T}_{21}$	$\mathcal{P} = \frac{1}{R_t}\mathcal{T}_{21}$	

Figure 3.3: Summary of Analogous Dynamic Elements

is nonlinear, due to the y^3-term. An equation like

$$\ddot{y} + y\dot{y} = u$$

is also non-linear as y and $\dot{y}$ are multiplied together.

Many non-linear systems can be linearised using Taylor-transforms.

5 Fundamentals and application of Laplace transforms

Laplace transforms are used frequently in Control theory to turn differential equations in the time-domain into algebraic equations in the Laplace-domain. In general, for a time signal $f(t)$, its Laplace transform is

$$F(s) = \mathscr{L}\left\{f(t)\right\} = \int_0^\infty f(t)e^{-st}\,\mathrm{d}t$$

where

$$s = \sigma + j\omega$$

is a complex variable.

To go from the Laplace-domain back to the time-domain the inverse Laplace transform, $\mathscr{L}^{-1}$ is used:

$$f(t) = \mathscr{L}^{-1}\{F(s)\} = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} F(s)e^{st}\,\mathrm{d}t.$$

A table of the most important Laplace Transform Pairs is given in Figure 5.1.

It is also worth noting that an integral in the time-domain corresponds to division by s in the Laplace-domain:

$$\mathscr{L}\left\{\int_0^\infty f(t)\,\mathrm{d}t\right\} = \frac{1}{s}\int_0^\infty f(t)e^{-st}\,\mathrm{d}t = \frac{1}{s}F(s)$$

$f(t)$	$F(s)$
Step function, $u(t)$	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
$\sin \omega t$	$\frac{\omega}{s^2+\omega^2}$
$\cos \omega t$	$\frac{s}{s^2+\omega^2}$
t^n	$\frac{n!}{s^{n+1}}$
$f^{(k)}(t) = \frac{d^k f(t)}{dt^k}$	$s^k F(s) - s^{k-1} f(0^-) - s^{k-2} f'(0^-) - \ldots - f^{(k-1)}(0^-)$
$\int_{-\infty}^{t} f(t)\,dt$	$\frac{F(s)}{s} + \frac{1}{s}\int_{-\infty}^{0} f(t)\,dt$
Impulse function $\delta(t)$	1
$e^{-at}\sin \omega t$	$\frac{\omega}{(s+a)^2+\omega^2}$
$e^{-at}\cos \omega t$	$\frac{s+a}{(s+a)^2+\omega^2}$
$\frac{1}{\omega}[(\alpha-a)^2+\omega^2]^{1/2} e^{-at}\sin(\omega t+\phi)$, $\phi = \tan^{-1}\frac{\omega}{\alpha-a}$	$\frac{s+\alpha}{(s+a)^2+\omega^2}$
$\frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin \omega_n\sqrt{1-\zeta^2}t, \zeta<1$	$\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$
$\frac{1}{a^2+\omega^2} + \frac{1}{\omega\sqrt{a^2+\omega^2}} e^{-at}\sin(\omega t-\phi)$, $\phi = \tan^{-1}\frac{\omega}{-a}$	$\frac{1}{s[(s+a)^2+\omega^2]}$
$1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}\sin\left(\omega_n\sqrt{1-\zeta^2}t+\phi\right)$, $\phi = \cos^{-1}\zeta, \zeta<1$	$\frac{\omega_n^2}{s(s^2+2\zeta\omega_n s+\omega_n^2)}$
$\frac{\alpha}{a^2+\omega^2} + \frac{1}{\omega}\left[\frac{(\alpha-a)^2+\omega^2}{a^2+\omega^2}\right]^{1/2} e^{-at}\sin(\omega t+\phi)$. $\phi = \tan^{-1}\frac{\omega}{\alpha-a} - \tan^{-1}\frac{\omega}{-a}$	$\frac{s+\alpha}{s[(s+a)^2+\omega^2]}$

Figure 5.1: Important Laplace Transform Pairs

and that a derivative in the time-domain corresponds to multiplication by s in the Laplace-domain

$$
\begin{aligned}
\mathscr{L}\left\{\frac{\mathrm{d}f(t)}{\mathrm{d}t}\right\} &= sF(s) - f(0)\\
\mathscr{L}\left\{\frac{\mathrm{d}f(t)^2}{\mathrm{d}t^2}\right\} &= s^2F(s) - sf(0) - f'(0)\\
\mathscr{L}\left\{\frac{\mathrm{d}f(t)^3}{\mathrm{d}t^3}\right\} &= s^3F(s) - s^2f(0) - sf'(0) - f''(0)\\
&\vdots\\
\mathscr{L}\left\{\frac{\mathrm{d}f(t)^n}{\mathrm{d}t^n}\right\} &= s^nF(s) - s^{n-1}f(0) - s^{n-2}f'(0) - \ldots - f^{(n-1)}(0).
\end{aligned}
$$

Further, addition/scaling of Laplace transforms follows the relation

$$\mathscr{L}\left\{af_1(t) \pm bf_2(t)\right\} = aF_1(s) \pm bF_2(s),$$

convolutions are defined as

$$\int_0^t f_1(t-\tau) f_2(\tau)\,\mathrm{d}\tau = F_1(s)F_2(s),$$

the initial-value theorem is

$$f(0+) = \lim_{s\to\infty} sF(s),$$

and the final-value theorem is

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s).$$

5.1 Transfer Functions

A transfer function is often used to describe the dynamics of a system. These are defined only for linear, stationary (i.e. constant parameter) systems, and may not apply for time-varying systems having one or mare time-varying parameters. A transfer function does not include any information about the internal structure of the system.

Definition 4: Transfer Function

A transfer function is defined as the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable, with all initial conditions assumed to be zero. I.e. the transfer function $G(s)$ of the Laplace transform $Y(s)$ of the output $y(t)$ and the Laplace transform $R(s)$ of the input $r(t)$ is

$$G(s) = \frac{Y(s)}{R(s)}.$$

Example: Transfer function of a Mass-spring-damper

For a mass-spring-damper such as that on Figure 3.1 we have Newton's second law:

$$M\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + b\frac{\mathrm{d}y}{\mathrm{d}t} + ky = r(t).$$

Now, taking the Laplace transform with zero initial conditions:

$$Ms^2Y(s) + bsY(s) + kY(s) = R(s) \implies Y(s)\left(Ms^2 + bs + k\right) = R(s).$$

So:

$$G(s) = \frac{Y(s)}{R(s)} = \frac{1}{Ms^2 + bs + k}.$$

In general for an n-th order linear differential equation:

$$\frac{\mathrm{d}^n y}{\mathrm{d}t^n} + q_{n-1}\frac{\mathrm{d}^{n-1}}{\mathrm{d}t^{n-1}} + \ldots + q_0 y = p_{n-1}\frac{\mathrm{d}^{n-1} r}{\mathrm{d}t^{n-1}} + p_{n-2}\frac{\mathrm{d}^{n-2} t}{\mathrm{d}t^{n-2}} + \ldots + p_0 r,$$

after taking the Laplace transform the transfer function will look like:

$$G(s) = \frac{Y(S)}{R(s)} = \frac{p_{n-1}s^{n-1} + p_{n-2}s^{n-2} + \ldots + p_0}{s^n + q_{n-1}s^{n-1} + \ldots + q_0} = \frac{p(s)}{q(s)}.$$

So in general, a transfer function is a ratio of two polynomials in s.

5.1.1 Poles and Zeros

From

$$G(s) = \frac{p(s)}{q(s)}$$

we define zeros to be the roots of $p(s) = 0$ and the poles to be the roots of $q(s) = 0$.

This means that zeros are values of s that make the transfer function become zero. These change how the system responds to input and affect transient behavior, frequency response, overshoot, etc.

The poles are more fundamental for system dynamics and determine stability, oscillation, decay rate and speed of responds. Poles with positive real part cause growth and instability as these lead to positive exponentials e^t after taking the inverse Laplace.

6 Partial Fraction Decomposition

Given a fraction such as

$$\frac{s+1}{(s+2)(s+3)}$$

we wish to split this into a sum of two fractions with simple denominators that we can easily take the inverse Laplace transform of

$$\frac{A}{s+2}+\frac{B}{s+3}$$

with some (yet) unknown constants A and B.

To do this we:

1. Expand into a term for each factor in the denominator
2. Recombine the RHS
3. Equate terms in s and constant terms, and have algebraic equations of the coefficients
4. Solve algebraic Equations
5. Each term is in a form so that inverse Laplace transforms can be applied.

Starting with:

$$\frac{s+1}{(s+2)(s+3)}=\frac{A}{s+2}+\frac{B}{s+3}.$$

We put the RHS over a common denominator:

$$\frac{A}{s+2}+\frac{B}{s+3}=\frac{A(s+3)+B(s+2)}{(s+2)(s+3)},$$

so now

$$\frac{s+1}{(s+2)(s+3)}=\frac{A(s+3)+B(s+2)}{(s+2)(s+3)}.$$

Since the denominators are equal, the numerators must also be equal:

$$\begin{aligned} s+1 &= A(s+3)+B(s+2) \\ &= As+3A+Bs+2B \\ &= (A+B)s+(3A+2B). \end{aligned}$$

From this we see that $A+B=1$ and $B=1-A$, so $A=-1$ and $B=2$. I.e.

$$\frac{s+1}{(s+2)(s+3)}=-\frac{1}{s+2}+\frac{2}{s+3}.$$

Now, taking the inverse Laplace transform is easy:

$$\mathscr{L}^{-1}\left\{\frac{1}{s+a}\right\}=e^{-at}$$
$$\mathscr{L}^{-1}\left\{\frac{s+1}{(s+2)(s+3)}\right\}=-e^{-2t}+2e^{-3t}.$$

Example: Partial Fraction Decomposition

$$
\begin{aligned}
G(s) &= \frac{5s+3}{(s+1)(s+2)(s+3)} \\
\frac{5s+3}{(s+1)(s+2)(s+3)} &= \frac{K_{s1}}{s+1} + \frac{K_{s2}}{s+2} + \frac{K_{s3}}{s+3} \\
K_{-1} &= \frac{p(-1)}{(-1+2)(-1+3)} = -1 \\
K_{-2} &= \frac{p(-2)}{(-2+1)(-2+3)} = 7 \\
K_{-3} &= \frac{p(-3)}{(-3+1)(-3+2)} = -6 \\
\frac{5s+3}{(s+1)(s+2)(s+3)} &= -\frac{1}{s+1} + \frac{7}{s+2} - \frac{6}{s+3} \\
G(t) &= -e^{-t} + 7e^{-2t} - 6e^{-3t}.
\end{aligned}
$$

Example: Conversion of ODE to Algebraic Equation using Laplace Transforms

Given the ODE

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y}{\mathrm{d}t} + 8y = 2 \qquad y(0) = y'(0) = 0$$

we want to find a solution using the Laplace transform to turn the above ODE into an easily-solvable algebraic equation.

We start by finding the Laplace transform of each term.

$$
\begin{aligned}
\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} &= s^2 Y(s) - sy(0) - y'(0) \\
6\frac{\mathrm{d}y}{\mathrm{d}t} &= 6\big(sY(s) - y(0)\big) \\
8y &= 8Y(s) \\
2 &= \frac{2}{s} \\
s^2 Y(s) - 0 - 0 + 6(sY(s) - 0) + 8Y(s) &= \frac{2}{s} \\
s^2 Y(s) + 6sY(s) + 8Y(s) &= \frac{2}{s} \\
Y(s)\left(s^2 + 6s + 8\right) &= \frac{2}{s} \\
Y(s) &= \frac{2}{s(s+2)(s+4)}.
\end{aligned}
$$

We have now found the solution in the Laplace-domain. To find the corresponding time-domain solution we take the inverse Laplace transform. To do this we use Partial Fraction Decomposition:

$$
\begin{aligned}
\frac{2}{s(s+2)(s+4)} &= \frac{K_{s1}}{s} + \frac{K_{s2}}{s+2} + \frac{K_{s3}}{s+4} \\
K_0 &= \frac{2}{2\cdot 4} = \frac{1}{4} \\
K_{-2} &= \frac{2}{-2\cdot 2} = -\frac{1}{2} \\
K_{-4} &= \frac{2}{-4\cdot -2} = \frac{1}{4} \\
\frac{2}{s(s+2)(s+4)} &= \frac{1}{4s} - \frac{1}{2(s+2)} + \frac{1}{4(s+4)}
\end{aligned}
$$

.

Now, that a partial fraction decomposition has been made we can easily take the inverse Laplace of the solution as

$$
\begin{aligned}
g(t) &= \frac{1}{4} - \frac{1}{2}e^{-2t} + \frac{1}{4}e^{-4t} \\
&= \frac{1}{4}e^{-4t}\left(e^{2t} - 1\right)^2 .
\end{aligned}
$$

Example: In-Class Exercise: Controlling a Laser Printer

A laser printer uses a laser beam to print copy rapidly for a computer. The laser is positioned by a control input $r(t)$, so that we have

$$Y(s) = \frac{4(s+50)}{s^2+30s+200}R(s).$$

The input $r(t)$ represents the desired position of the laser beam.

(a) If $r(t)$ is a unit step input find the output $y(t)$.

(b) What is the final value of $y(t)$?

(a) For a unit step $r(t)$, $R(s) = \frac{1}{s}$, so

$$
\begin{aligned}
Y(s) &= \frac{4(s+50)}{s^2+30s+200}\frac{1}{s} \\
&= \frac{4(s+50)}{s^3+30s^2+200s} \\
&= \frac{4(s+50)}{s(s+10)(s+20)} \\
\frac{4(s+50)}{s(s+10)(s+20)} &= \frac{K_0}{s} + \frac{K_{10}}{s+10} + \frac{K_{20}}{s+20} \\
K_0 &= \frac{200}{10\cdot 20} = 1 \\
K_{-10} &= \frac{160}{-10\cdot 10} = -\frac{8}{5} \\
K_{-20} &= \frac{120}{-20\cdot -10} = \frac{3}{5} \\
Y(s) &= \frac{1}{s} - \frac{8}{5(s+10)} + \frac{3}{5(s+20)} \\
y(t) &= 1 - \frac{8}{5}e^{-10t} + \frac{3}{5}e^{-20t}.
\end{aligned}
$$

(b) We solve:

$$\lim_{t\to\infty} 1 - \frac{8}{5}e^{-10t} + \frac{3}{5}e^{-20t} = 1.$$

Lecture 2: Mathematical Models of Systems

20. Marts 2026

6.1 Block Diagrams

Though mathematical models are useful in describing the complex behavior of a control system they are often rather unintuitive. A more graphical alternative to the rigorous mathematical models are Block Diagrams. An example of one of these is shown on Figure 6.1

Definition 5: Block Diagram

A *Block Diagram* is a schematic representation of the cause-and-effect relationshop throughout the system together with the transfer functions of the variables of interest.

Typical elements of block diagrams include but are not limited to:

- Comparators
- Blocks representing individual component transfer functions (sensors, actuators, controllers, plants, etc.)
- Input or reference signals
- Output signals
- Disturbance signals
- Feedback loops

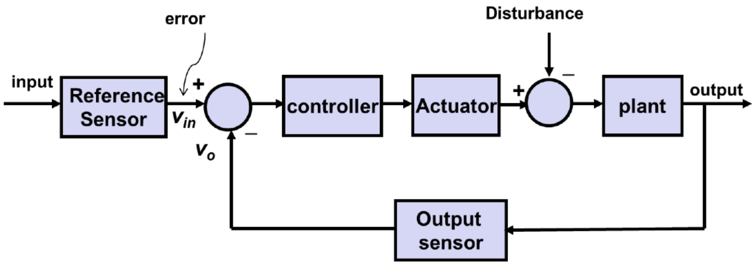


Figure 6.1: Example of a block diagram.

An example showing how different components of a system might be represented as blocks of a block diagram is shown in Figure 6.2.

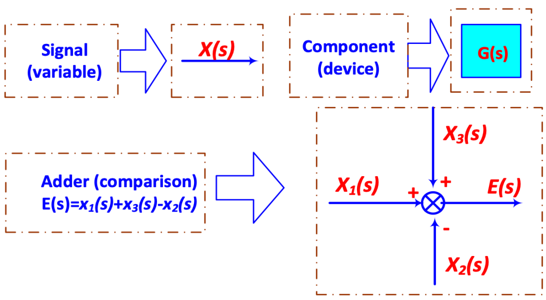


Figure 6.2: Different components represented as blocks.

For multiple input or output systems the subscript convention is such that G_{ij} is the transfer function of output i and input j.

6.1.1 Transformation and Reduction of Block Diagrams

Different Block diagrams can be transformed and reduced into more simple forms. This is especially useful for large and complex systems.

For a simple cascade block diagram as shown on Figure 6.3 one is able to reduce the block diagram into a single $G_1 \cdot G_2$-block.

Figure 6.3: Cascade Block diagram.

For a parallel block diagram as shown on Figure 6.4 the resulting transformation is a single $G_1 - G_2$-block.

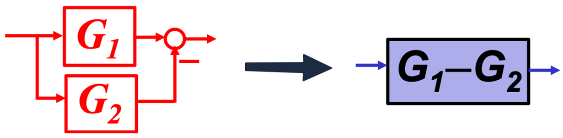


Figure 6.4: Parallel block diagram.

For a negative feedback Block diagram as shown on Figure 6.5 the reduction is a single block:

$$\frac{G_1}{1 + G_1 G_2}.$$

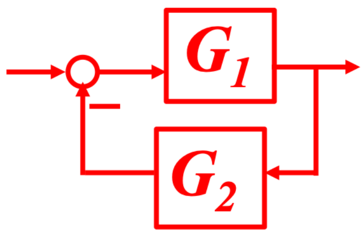


Figure 6.5: Feedback block diagram.

A summing node is able to be moved across a block as shown in Figure 6.6.

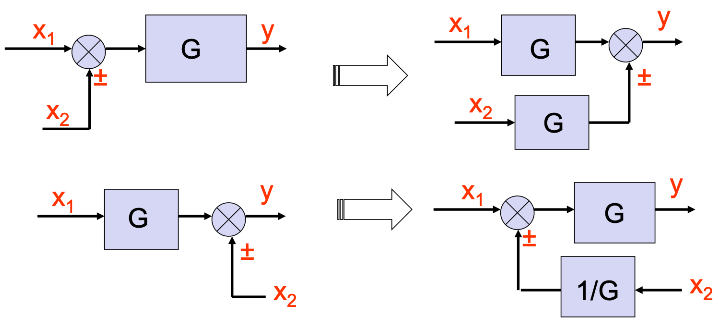


Figure 6.6: Movement of a summing node across a block.

Pickoff points can also be moved across blocks. The manner this is done by is shown in Figure 6.7.

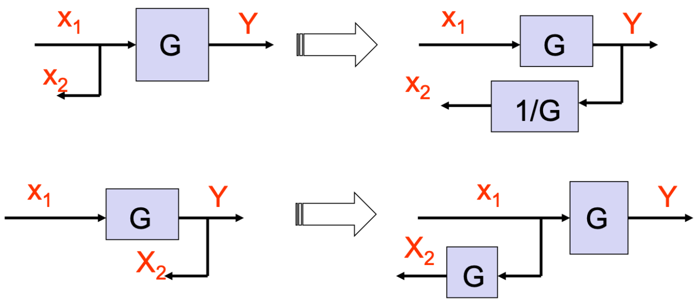


Figure 6.7: Movement of a pickoff point across a block.

It is also possible to interchange neighboring summing points or pickoff points as shown in Figure 6.8.

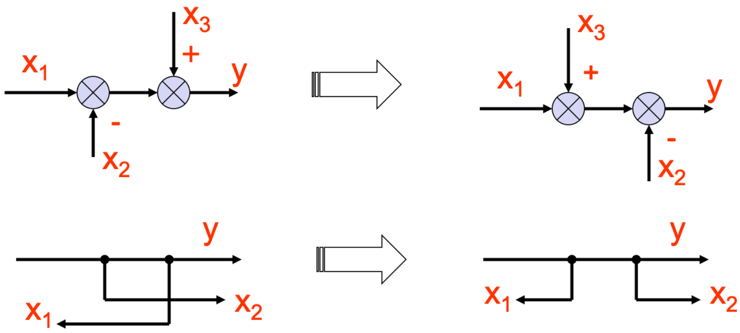


Figure 6.8: Interchanging neighboring summing and pickoff points

Example: Transfer function of an interacting system

Given the system shown find the transfer function.

We combine all the feedback loops using:

$$G_i(s)||H_i(s) = \frac{G_i(s)}{1 - G_i(s)H_i(s)}.$$

This gives:

$$
\begin{aligned}
G_2(s)||H_2(s) &= \frac{G_2(s)}{1-G_2(s)H_2(s)}\\
G_3(s)||H_3(s) &= \frac{G_3(s)}{1-G_3(s)H_3(s)}\\
G_6(s)||H_6(s) &= \frac{G_6(s)}{1-G_6(s)H_6(s)}\\
G_7(s)||H_7(s) &= \frac{G_7(s)}{1-G_7(s)H_7(s)}.
\end{aligned}
$$

Now the two branches are both cascading, so using G_1G_2 for series we get:

$$
\begin{aligned}
B_1(s) &= G_1(s)\cdot\frac{G_2(s)}{1-G_2(s)H_2(s)}\cdot\frac{G_3(s)}{1-G_3(s)H_3(s)}\cdot G_4(s)\\
B_2(s) &= G_5(s)\cdot\frac{G_6(s)}{1-G_6(s)H_6(s)}\cdot\frac{G_7(s)}{1-G_7(s)H_7(s)}\cdot G_8(s).
\end{aligned}
$$

Now, we are left with the two parallel $B_1(s)$ and $B_2(s)$ branches which are combined using $G_1||G_2 = G_1+G_2$ as:

$$B_1(s)+B_2(s) = G_1(s)\cdot\frac{G_2(s)}{1-G_2(s)H_2(s)}\cdot\frac{G_3(s)}{1-G_3(s)H_3(s)}\cdot G_4(s)+G_5(s)\cdot\frac{G_6(s)}{1-G_6(s)H_6(s)}\cdot\frac{G_7(s)}{1-G_7(s)H_7(s)}\cdot G_8(s).$$

Our transfer function is therefore:

$$t(s) = G_1(s)\cdot\frac{G_2(s)}{1-G_2(s)H_2(s)}\cdot\frac{G_3(s)}{1-G_3(s)H_3(s)}\cdot G_4(s)+G_5(s)\cdot\frac{G_6(s)}{1-G_6(s)H_6(s)}\cdot\frac{G_7(s)}{1-G_7(s)H_7(s)}\cdot G_8(s).$$

So

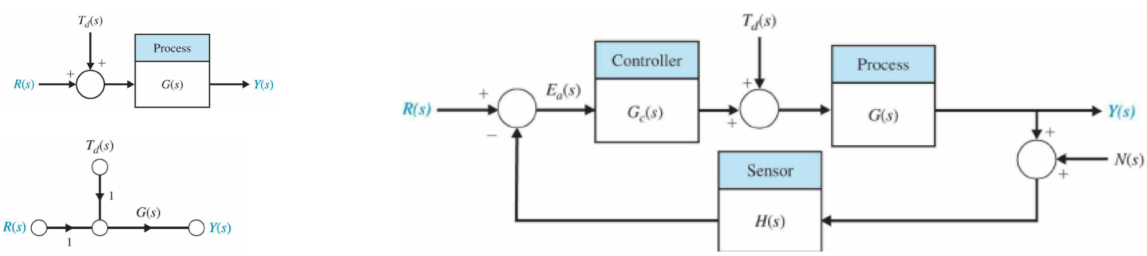
$$Y(s) = R(s)\left(G_1(s)\cdot\frac{G_2(s)}{1-G_2(s)H_2(s)}\cdot\frac{G_3(s)}{1-G_3(s)H_3(s)}\cdot G_4(s)+G_5(s)\cdot\frac{G_6(s)}{1-G_6(s)H_6(s)}\cdot\frac{G_7(s)}{1-G_7(s)H_7(s)}\cdot G_8(s)\right)$$

Lecture 3: Feedback Control System Characteristics 24. March 2026t

7 Feedback Control System Characteristics

7.1 Basic Concepts and Structure

Whereas an open-loop control system operates without feedback (see Figure 7.1a, a closed-loop system (Figure 7.1b) uses a measurement of the outout signal and a comparison with the desired output to generate an error signal that is used by the controller to adjust the actuator, which has certain robustness against disturbances and parameter changes.



(a) Open-loop control system.

(b) Closed-loop control system.

Figure 7.1: Comparison of an open and closed loop control system.

7.2 Error Signal Analysis

On Figure 7.2 we define the tracking error $E(s)$ to be

$$E(s) = R(s) - Y(s)$$

because with feedback $H(s) = 1$, the measured output is just the output itself. I.e. if $Y(S)$ follows $R(s)$ perfectly, then $E(s) = 0$. On Figure 7.2 the system is influenced by three inputs the desired reference $R(s)$, the disturbance $T_d(s)$ and the measurement noise $N(s)$.

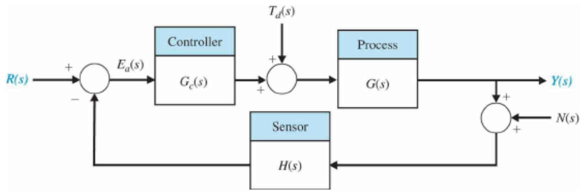


Figure 7.2: Tracking error with unity feedback, $H(s) = 1$.

With feedback $H(s) = 1$, the output $Y(s)$ can be computed as a sum of three contributions. One from the reference $R(s)$, which is the usual closed-loop transfer function from reference to output:

$$Y_R = \frac{G_c G}{1 + G_c G} R,$$

one from the disturbance $T_d(s)$, which shows how much the disturbance affects the output

$$Y_T = \frac{G}{1 + G_c G} T_D,$$

and one from the measurement noise

$$Y_N = -\frac{G_c G}{1 + G_c G} N.$$

Note the minus sign in the above appears as the noise is in the feedback signal, thus corrupting the measured output and thereby altering the error in the opposite direction. Which means the output is

$$Y(s) = Y_R + Y_T + Y_N = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} R(s) + \frac{G(s)}{1 + G_c(s)G(s)} T_D(s) - \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} N(s).$$

The error $E(s)$ is then

$$E(s) = R(s) - Y(s) = \frac{1}{1 + G_c(s)G(s)} R(s) - \frac{G(s)}{1 + G_c(s)G(s)} T_d(s) + \frac{G_c(s)G(s)}{1 + G_c(s)G(s)} N(s).$$

We now define the loop gain $L(s)$ as the total gain you get by going around a feedback loop

$$L(s) = G_c(s)G(s)H(s)$$

and since $H(s) = 1$,

$$L(s) = G_c(s)G(s).$$

This simplifies the formula for the error to:

$$E(s) = \frac{1}{1 + L(s)} R(s) - \frac{G(s)}{1 + L(s)} T_d(s) + \frac{L(s)}{1 + L(s)} N(s).$$

If we now define the sensitivity function $S(s)$ as

$$S(s) = \frac{1}{1 + L(s)}$$

and the complementary sensitivity function $C(s)$ as

$$C(s) = \frac{L(s)}{1 + L(s)}$$

then the error becomes

$$E(s) = S(s)R(s) - S(s)G(S)T_d(s) + C(s)N(s).$$

Here, the sensitivity function $S(s) = 1/(1 + L(s))$ tells how much of the reference mismatch and disturbance gets through. From the formula we see that large $L(s)$ leads to small $S(s)$ so large loop gain reduces tracking error $R(s)$ and the effect of disturbance $T_d(s)$ On the other hand, the complementary sensitivity $C(s) = L(s)/(1 + L(s))$ tell how much measurement noise gets through into the error/output. If $L(s)$ is large, then $C(s) \approx 1$, hence high gain is bad for noise sensitivity. Further, we know that

$$S(s) + C(s) = 1$$

hence both cannot be small at the same frequency. Making $S(s)$ small helps tracking and disturbance rejection and making $C(s)$ small helps noise rejection, but improving one tends to worsen the other.

Generally, disturbances are more important at low frequencies and measurement noise is usually more important at high frequencies. Therefore, a good controller typically will try to make the loop gain $L(s)$ large at low frequencies and smaller at higher frequencies. At low frequencies, if $L(s) \gg 1$, then

$$S(s) \approx 0, \qquad C(s) \approx 1$$

so the tracking error and disturbance effects are small. At high frequencies, if $L(s) \ll 1$, then

$$S(s) \approx 1, \qquad C(s) \approx 0$$

so measurement noise is suppressed. I.e. We want the controller to fight slow changes strongly, but ignore very fast noisy fluctuations.

7.3 Sensitivity to Parameter Variations

The input $G(s)$ is time-varying in nature due to uncertainties such as changing environments, aging, and other factors. These uncertainties will, for open-loop systems result in inaccurate output or low performance, however this can be mitigated in a closed-loop system. As the process $G(s)$ undergoes a change $\Delta G(s)$ a resulting change in the tracking error $\Delta E(s)$ will occur. For an open-loop system this will be

$$\Delta E(s) = \Delta G(s) G_c(s) R(s)$$

and for a closed loop system it will be:

$$\begin{aligned}\Delta E(s) &= \frac{-G_c(s)\Delta G(s)}{(1+G_c(s)G(s)+G_c(S)\Delta G(s))\,(1+G_c(s)G(s))}R(s)\\ &\approx \frac{-G_c(s)\Delta G(s)}{(1+G_c(s)G(s))^2}R(s)\\ &= \frac{-G_c(s)\Delta G(s)}{(1+L(s))^2}R(s)\\ &\approx -\frac{1}{L(s)}\frac{\Delta G(s)}{G(s)}R(s).\end{aligned}$$

I.e. for a closed-loop system the error change caused by plant uncertainties is proportional to the relative plant change $\Delta G/G$ divided by the loop gain $L(s)$. So bigger changes in input result in bigger changes in output but a bigger loop gain results in a smaller error change. If e.g. $|L(s)| = 100$, then the effect of plant variation is suppressed by about a factor 100 at that frequency.

7.3.1 Sensitivity

We can define system sensitivity S as

$$S = \frac{\Delta T(s)/T(s)}{\Delta G(s)/G(s)}$$

where

$$T(s) = \frac{Y(s)}{R(s)}$$

is the closed-loop transfer function from reference to output. Sensitivity tells how much the transfer function changes, in percentage, when the plant changes by a percentage amount. So if $S = 0{,}1$, then a 10% change in G causes only about a 1% change in T. This also means that a smaller sensitivity is generally better than a higher one.

For very small changes, percent changes become derivatives:

$$S = \frac{\partial T(s)/T(s)}{\partial G(s)/G(s)} = \frac{\partial \ln T}{\partial \ln G}$$

which is the infinitesimal version of the definition of sensitivity.

We also define S_G^T as the sensitivity of T with respect to G. For an open-loop system, then

$$T(s) = G_c(s)G(s) \implies S_G^T = 1$$

as T is directly proportional to G. Meaning a 1% change in G results in a 1% change in the system transfer function T, so the loop is fully sensitive. For a closed-loop with unity feedback

$$T(s) = \frac{G_c(s)G(s)}{1+G_c(s)G(s)} \implies S_G^T = \frac{1}{1+G_c(s)G(s)} = \frac{1}{1+L(s)}$$

which is exactly the same as the usual sensitivity function.

Instead of asking how much T changes with the whole plant G, we can also ask how T changes with some parameter α such as mass, damping, stiffness, time constant, gain constant, etc. Then:

$$S_\alpha^T = S_G^T S_\alpha^G$$

following the chain-rule. This can be read as the first parameter, α changes the plant G which then changes the system transfer T. I.e. total sensitivity from α to T is

$$(\text{sensitivity of } T \text{ to } G) \cdot \big(\text{sensitivity of } G \text{ to } \alpha\big).$$

We can also write

$$S_\alpha^T = \frac{\partial \ln T}{\partial \ln \alpha}$$

and if

$$T(s,\alpha) = \frac{N(s,\alpha)}{D(s,\alpha)}$$

then

$$S_\alpha^T = \frac{\partial \ln N}{\partial \ln \alpha} - \frac{\partial \ln D}{\partial \ln \alpha} = S_\alpha^N - S_\alpha^T.$$

7.4 Disturbance Rejection in Feedback Systems

A disturbance signal is an unwanted signal that affects the output signal. From

$$E(s) = \frac{1}{1+L(s)}R(s) - \frac{G(s)}{1+L(s)}T_d(s) + \frac{L(s)}{1+L(s)}N(s)$$

with $R(s) = N(s) = 0$ we see the disturbance contribution is

$$E_T(s) = -S(s)G(s)T_d(s) = -\frac{G(s)}{1+L(s)}T_d(s)$$

as mentioned before, large loop gain $L(s) = G_c(s)G(s)$ leads to good disturbance rejection, and disturbance signals are often low frequency so it is desirable to design $G_c(s)$ such that $S(s)$ is small at low frequencies.

To imagine how disturbance arises imagine e.g. a steel bar being moved and rolled by a motor trying to maintain constant speed for the steel bar. However when the load changes (e.g. due to adding weight to the bar, or drying out of lubricant) the required torque changes too. That change acts like a disturbance torque T_d. In this scenario, if the controller command is unchanged the rotational speed ω can drop.

7.4.1 Open-loop Speed Under Disturbance

We now want to determine what the speed change of a motor might be due to load disturbance. For the open-loop motor model on Figure 7.4, the error becomes

$$E(s) = -\omega(s) = \frac{1}{Js + b + \frac{K_m K_b}{R_a}}T_d(s).$$

The sign here is negative because a positive load torque opposes motion, so it reduces speed.

If the disturbance is a step torque

$$T_d(s) = \frac{D}{s}$$

then by the final value theorem

$$\lim_{t\to\infty} E(t) = \lim_{s\to 0} sE(s)$$

which gives

$$E(\infty) = \frac{D}{b + \frac{K_m K_b}{R_a}}$$

and since $E = -\omega$

$$\omega(\infty) = -\frac{D}{b + \frac{K_m K_b}{R_a}}.$$

This means that a constant extra load torque creates a constant steady speed drop. I.e. in open-loop, the motor cannot fully reject the step disturbance, it just settles to a lower speed.

7.4.2 Closed-loop Speed Under Disturbance

Now, we introduce a feedback controller to reduce the loading disturbance effect as shown on Figure 7.3. Here, we also reorganize the system as shown on Figure 7.3, into

$$G_1(s) = \frac{K_a K_m}{R_a}, \qquad G_2(s) = \frac{1}{Js+b}, \qquad H(s) = K_t + \frac{K_b}{K_a}.$$

The disturbance $T_d(s)$ is injected between G_1 and G_2 and the output is fed back through $H(s)$.

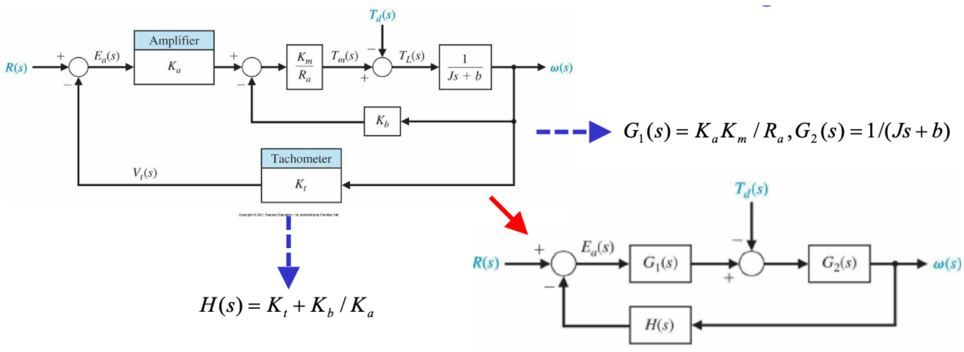


Figure 7.3: Closed-loop model.

For disturbance only, the steady-state error expression for the closed-loop system becomes:

$$E(s) = -\omega(s) = \frac{G_2(s)}{1 + G_1(s)G_2(s)H(s)} T_d(s)$$

which has the same structure as before:

$$\text{disturbance effect} = \frac{\text{plant term}}{1 + \text{loop gain}}.$$

If

$$G_1(s)G_2(s)H(S) \gg 1$$

then

$$E(s) \approx \frac{1}{G_1(s)H(s)} T_d(s)$$

i.e. for large feedback the disturbance is not much dominated by G_2; it is instead suppressed roughly in inverse proportion to the loop gain. Also

$$G_1(s)H(s) = \frac{K_a K_m}{R_a}\left(K_t + \frac{K_b}{K_a}\right)$$

and for $K_a \gg K_b$

$$G_1(s)H(s) \approx \frac{K_1 K_m K_t}{R_a}$$

so increasing amplifier gain K_a improves disturbance rejection.

With $R(s) = 0$, the closed-loop speed under disturbance becomes

$$\omega_c(s) = -\frac{1}{Js + b + \frac{K_m}{R_a}(K_t K_a + K_b)} T_d(s).$$

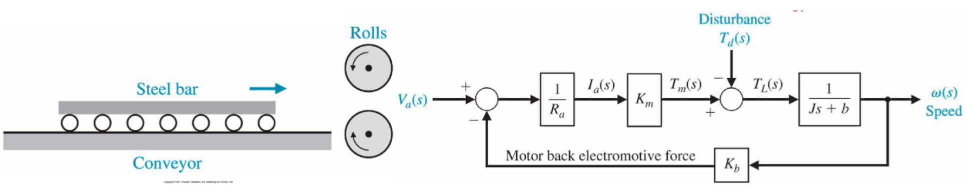


Figure 7.4: Open-loop motor model.

Comparing this with the open-loop version

$$\omega_o(s) = -\frac{1}{Js + b + \frac{K_m K_b}{R_a}} T_d(s)$$

we notice that the closed-loop version has an extra stabilizing term $K_t K_a K_m / R_a$ in the denominator.

For a step disturbance $T_d(s) = D/s$, the final value theorem gives

$$\omega(\infty) = -\frac{D}{b + \frac{K_m}{R_a}(K_t K_a + K_b)}$$

for sufficiently large K_a, this is

$$\omega_c(\omega) \approx -\frac{R_a}{K_a K_m K_t} D.$$

Meaning the steady speed drop is now roughly proportional to $1/K_a$.

Taking the quotient ω_c/ω_o we get

$$\frac{\omega_c(\infty)}{\omega_o(\infty)} = \frac{R_a b + K_m K_b}{K_a K_m K_t}$$

which is usually less than 0,02. That means the disturbance effect in closed loop systems can be less than 2% of what it would be in open loop.

7.5 Measurement Noise Attenuation

Measurement noise is noise added in the sensor or measurement path. The plant output may be fine, but the controller does not see the true output. Instead it sees

$$y_{\text{sensor}} = y + n$$

there n is the noise. That noise is then fed back, so the controller may react to the noise as if it were a real signal. Only looking at the noise contribution to the error we see

$$E_n(s) = C(s)N(s) = \frac{L(s)}{1 + L(s)} N(s)$$

where $C(s) = L(s)/(1 + L(s))$ is the complementary sensitivity function. This is complementary to the sensitivity function from above and a trade-off between them is therefore made at a given frequency.

7.6 The Transient Response Control

The transient response is the response of a system as a function of time, and must be adjusted until it is satisfactory. For an open-loop system, the process $G(s)$ must be replaces with a more suitable process, or design the cascade controller $G_c(s)$. For a closed-loop system, feedback loop parameters can often be adjusted to yield the desired response.

7.7 Steady-State Error

The steady state error is the error after the transient response has decayed, leaving only the continuous response.

$$e_{ss} = \lim_{t\to\infty} e(t).$$

In an open loop, the output is just

$$Y(s) = G(s)R(s)$$

so the error is

$$E_o(s) = R(s) - Y(s) = (1 - G(s))R(s)$$

for a unit step input, $R(s) = 1/s$, so

$$E_o(s) = (1 - G(s))\frac{1}{s}.$$

Now, applying the final value theorem we get:

$$e_o(\infty) = \lim_{s\to 0} sE_o(s) = \lim_{s\to 0} s(1 - G(s))\frac{1}{s} = 1 - G(0).$$

I.e. the steady-state error depends on the DC gain $G(0)$. If $G(0) = 1$, then $e_o(\infty) = 0$, however $G(0) \neq 1$ leads to a nonzero steady state error.

For a closed loop system, the feedback means a much smaller error is produced. For unity feedback $H(s) = 1$, the error transfer function is

$$E_c(s) = \frac{1}{1 + G_c(s)G(s)}R(S).$$

For a unit step input

$$E_c(s) = \frac{1}{1 + G_c(s)G(s)}\frac{1}{s}.$$

Now, applying the final value theorem, we get

$$e_c(\infty) = \lim_{s\to 0} sE_c(s) = \lim_{s\to 0} s\left(\frac{1}{1 + G_c(s)G(s)}\frac{1}{s}\right)$$

so

$$e_c(\infty) = \frac{1}{1 + G_c(0)G(0)}$$

or using the DC loop gain:

$$e_c(\infty) = \frac{1}{1 + L(0)}.$$

So if the DC loop gain is large, then the steady-state error is small.

7.8 The Cost of Feedback

In general, feedback is very useful to make more accurate systems, however one must always keep the cost of adding said feedback in mind before deciding if it is worth the investment. Adding feedback leads to increased complexity, as sensors, which are expensive, noisy, and inaccurate, need to be added. The closed loop gain of a feedback system will also decrease by a factor of $1/(1 + G_c(s)G(s))$. Further, feedback introduces possible instability of the system, which does not arise for open-loop systems.

Lecture 4: Performance of Feedback Control Systems 27. March 2026

8 Performance of Feedback Control Systems

To really understand the following part on Performance of Feedback Control Systems, a few concepts must first be defined.

Definition 6: Transient Response

The part of the response that disappears with time.

Definition 7: Steady-State Response

The part of the response that exists for a long time following an input signal initiation.

Definition 8: Design Specifications

The measures of performance indicating the quality of the system to the designer.

8.1 Standard Test Input Signals

In control theory, we seldom know the exact real-world input a system will see. Instead, we opt to test the system using simple "benchmark" inputs whose responses are easy to analyze. This helps us answer a lot general questions about the response of a system. A few of the most common standard test input signals are:

- Step Input
- Ramp Input
- Parabolic Input
- Sinusoidal Input
- Impulse Input

The three most common time-domain test inputs are the step-input, the ramp-input, and the parabolic inputs. These are all depicted on Figure 8.1.

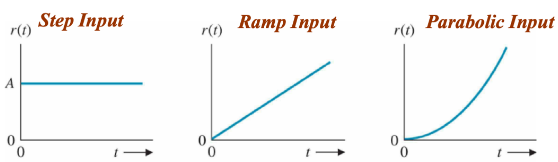


Figure 8.1: Common input signals.

8.1.1 Step Input

A step input jumps suddenly from 0 to a constant value A

$$r(t) = A, t > 0$$

and its Laplace transform is

$$R(s) = \frac{A}{s}.$$

This corresponds to a sudden physical command. Such as setting a motor to a high signal.

8.1.2 Ramp Input

A ramp input grows linearly with time

$$r(t) = At, t > 0$$

and its Laplace transform is

$$R(s) = \frac{A}{s^2}.$$

This corresponds to a steadily changing command, such as asking a vehicle to move forward at constant speed in position coordinates.

8.1.3 Parabolic Input

A parabolic input grows proportionally to t^2

$$r(t) = At^2, t > 0$$

and its Laplace transform is

$$R(s) = \frac{2A}{s^3}.$$

This corresponds to a constantly accelerating command, such as a position reference under constant acceleration.

Worth noting is that the Ramp Input is simply the integral of the Step Input; and the Parabolic Input is the integral of the Ramp Input.

8.1.4 Unit Impulse Response

The unit impulse response is defined with regard to the impulse $\delta(t)$. The ideal impulse is infinitely narrow and infinitely tall, but has area 1 (like an infinitely thin standard normal distribution). We define this approximately as:

$$\delta(t) = \begin{cases} \frac{1}{\epsilon}, & -\frac{\epsilon}{2} \leq t \leq \frac{\epsilon}{2} \\ 0, & \text{otherwise} \end{cases}$$

as $\epsilon \to 0$.

If the transfer function is $G(s)$, then the system's impulse response is:

$$g(t) = \mathscr{L}^{-1}\{G(s)\}.$$

This is an extremely important result, as it states that for an LTI system, the impulse response completely characterizes the system.

A convolution is defined as:

$$y(t) = \int_{-\infty}^{t} g(t-\tau)r(\tau)\,\mathrm{d}\tau.$$

I.e. the output is the convolution of the input $r(t)$ with the impulse response $g(t)$. In other words, any input can be built from many tiny impulses, and the output is the sum of the system's response to all those tiny impulses. In Laplace form this is

$$Y(s) = G(s)R(s)$$

which is the transform-domain version of convolution.

8.1.5 Sinusoidal Input

The sinusoidal input is defined by

$$r(t) = \begin{cases} A\sin\omega t, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

with Laplace transform:

$$R(s) = \frac{A\omega}{s^2+\omega^2}.$$

8.1.6 Standard Test Signal Input

In general, step, ramp, and parabolic inputs can be written as

$$r(t) = t^n$$

with Laplace transform

$$R(s) = \frac{n!}{s^{n+1}}$$

8.2 Performance of Second-Order Systems

We consider the general second-order system on Figure 8.2. Here, we have the open-loop transfer function

$$G(s) = \frac{\omega_n^2}{s(s+2\zeta\omega_n}$$

with unity negative feedback. This results in the closed-loop transfer function becoming

$$\frac{Y(s)}{R(s)} = \frac{G(s)}{1+G(s)} = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}.$$

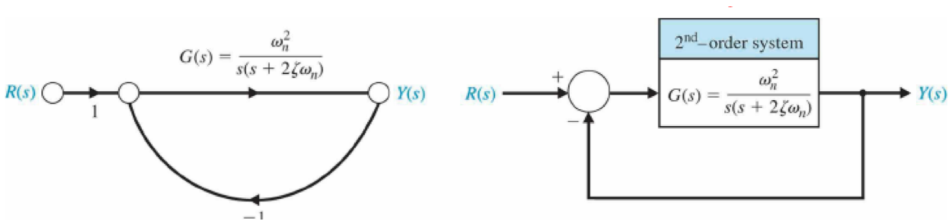


Figure 8.2: Second order system.

Now, we provide the system with a unit step input

$$R(s) = \frac{1}{s}$$

so

$$Y(s) = \frac{\omega_n^2}{s\left(s^2+2\zeta\omega_n s+\omega_n^2\right)}$$

and in time domain, for the underdamped case $0 < \zeta < 1$,

$$y(t) = 1 - \frac{1}{\beta}e^{-\zeta\omega_n t}\sin\left(\omega_n\beta t+\theta\right)$$

where

$$\beta = \sqrt{1-\zeta^2}, \qquad \theta = \cos^{-1}(\zeta).$$

This corresponds to a final value of 1 plus an oscillation whose amplitude decays with the exponential term. The factor

$$e^{-\zeta\omega_n t}$$

tells how fast the oscillation dies out, so larger ζ results in faster decay and less overshoot, whilst smaller ζ results in slower decay and more ringing.

For the above case, the poles are

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}.$$

Here, $-\zeta\omega_n$ determines how fast the oscillations die out; a more negative real part results in faster decay, and a real part closer to 0 results in slower decay. The imaginary part

$$\omega_n\sqrt{1-\zeta^2}$$

determines the oscillation frequency, i.e. it is the damped natural frequency

$$\omega_d = \omega_n\sqrt{1-\zeta^2}.$$

The pole plots on Figure 8.3, show how the roots move as ζ changes.

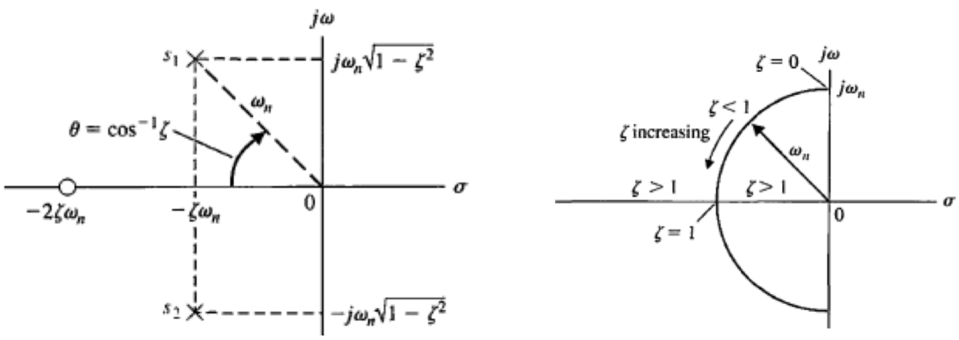


Figure 8.3: Pole plots

When $0 < \zeta < 1$, the system is underdamped, meaning the poles are complex conjugates

$$-\zeta\omega_n \pm j\omega d.$$

When $\zeta = 1$ the system is critically damped. Here, the two poles meet on the real axis.

When $\zeta > 1$, the system is overdamped. Here, the poles are real and distinct, and the responds has no oscillation.

For $\zeta = 0$, the system is undamped. This gives poles on the imaginary axis $\pm j\omega_n$, which correspond to sustained oscillation with no decay.

Also on Figure 8.3, a geometric relationship in the s-plane is shown. For poles of a standard second-order system the distance from the origin is ω_n and the angle is related to ζ by

$$\cos\theta = \zeta$$

so for fixed ω_n, changing ζ moves the poles along a circle of radius ω_n.

On Figure 8.4 the step time response is shown for different damping ratios. For the small damping ratios, we observe fast initial rise, large overshoot, and strong oscillation. For the medium damping ratio, we have some overshoot, and less oscillation. For critical damping, we have no oscillation, and no overshoot. For the overdamped case, we have no overshoot, and no overshoot, but with a sluggish response.

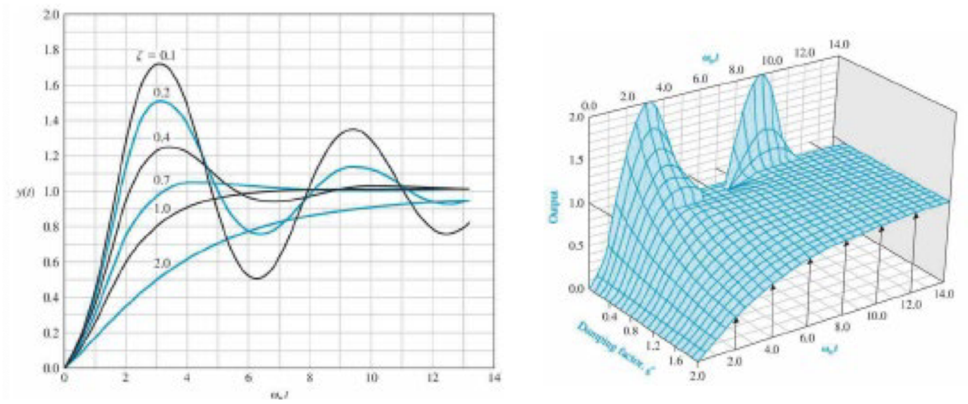


Figure 8.4: Time responses for different damping ratios

In general, we can observe that as the damping ratio decreases, the closed-loop roots approach the imaginary axis, and the response becomes increasingly oscillatory.

8.3 Performance Specification Definition

On Figure 8.5 a typical step response $y(t)$ approaching its final value $y(\infty)$ is depicted.

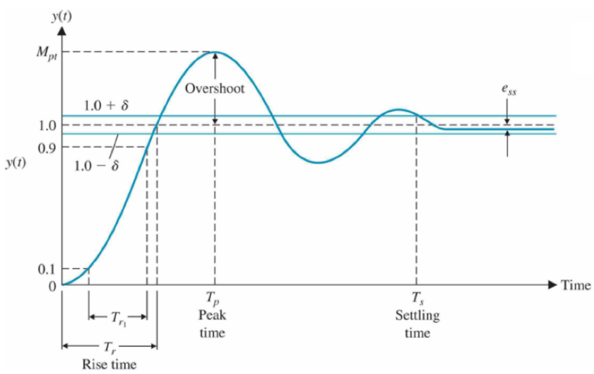


Figure 8.5: Typical step response $y(t)$ going toward a final value $y(\infty)$.

8.3.1 Rise Time

We define the rise time on Figure 8.5 as the time it takes the output to "rise" to its target value. For an underdamped system ($\zeta < 1$), the rise time T_r is the first time the response reaches the final value $y(\infty)$. For an overdamped system ($\zeta > 1$), the response never overshoots, so rise time is often defined as the 10% to 90% time:

$$T_{r1} = T_2 - T_1$$

with

$$y(T_1) = 0{,}1\,y(\infty), \qquad y(T_2) = 0{,}9\,y(\infty).$$

8.3.2 Peak Time

The peak time on Figure 8.5, is defined as the time, when the response reaches its first maximum. At the peak, the slope is zero

$$\left.\frac{\mathrm{d}y(t)}{\mathrm{d}t}\right|_{t=T_p} = 0.$$

This metric only makes sense for an underdamped response, because overdamped responses do not have a peak above the final value.

8.3.3 Percent Overshoot

On Figure 8.5, the percent overshoot is defined as

$$\sigma_p\% = \frac{y(T_p) - y(\infty)}{y(\infty)} \cdot 100\%.$$

I.e. how much higher than the final value did the first peak go, as a percentage.

For the standard second-order system

$$\%\mathrm{OS} = 100 e^{-\zeta\pi\frac{1}{\sqrt{1-\zeta^2}}}, \qquad 0 < \zeta < 1)$$

I.e. small ζ leads to lots of oscillation and a large overshoot and vice versa.

8.3.4 Settling Time

For an underdamped second-order system, the oscillations decay with an exponential envelope like

$$e^{-\zeta\omega_n t}.$$

A few different conventions exist for the settling time, the two most common of which being the 2% criterion and the 5% criterion. If you want your response to be within 2%, you use

$$e^{-\zeta\omega_n T_s} < 0{,}02$$

which gives the approximation

$$T_s \approx \frac{4}{\zeta\omega_n} = 3\tau$$

for the 2% criterion, with $\tau = 1/(\zeta\omega_n)$ being the exponential decay time constant.

Likewise, for the 5% criterion, a common approximation is

$$T_s \approx \frac{3}{\zeta\omega_n} = 3\tau.$$

8.3.5 Delay Time

The delay time, T_d on Figure 8.5, is defined by

$$y(T_d) = 0{,}5\, y(\infty)$$

so T_d is the time it takes the output to reach 50% of its final value.

8.3.6 Performance Compromise

Two important formulas from the above is the formula for the peak time and for the percentage overshoot:

$$T_P = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} \qquad\qquad \%\mathrm{OS} = 100 e^{-\zeta\pi\frac{1}{\sqrt{1-\zeta^2}}}.$$

These are depicted on Figure 8.6. Here we see that, as the damping ratio ζ increases the percent overshoot decreases and the normalized peak time $\omega_n T_p$ increases, so more damping leads to less overshoot but also a slower response.

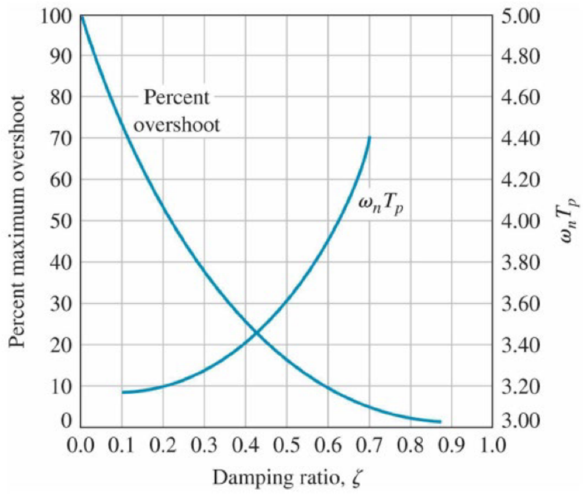


Figure 8.6: Performance compromise of control systems.

This is why a response cannot be made both extremely fast and extremely smooth with no overshoot without changing other elements of the design.

8.3.7 Swiftness of the System

On Figure 8.7 the normalized rise time $\omega_n T_{r1}$ is plotted versus the damping ratio ζ. Also present, is a linear approximation of T_{r1} stating:

$$T_{r1} \approx \frac{2{,}16\zeta + 0{,}60}{\omega_n}$$

I.e. increasing ζ increases rise time and increasing ω_n decreases rise time.

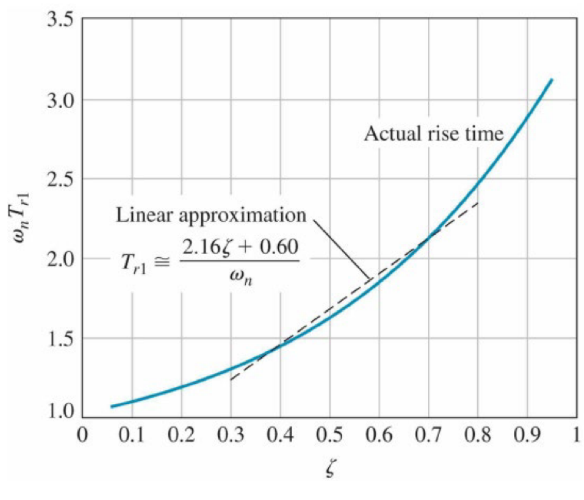


Figure 8.7: Normalized rise time $\omega_n T_{r1}$ versus ζ with linear approximation.

8.4 Effect of a Third Pole

We consider a third-order system with a closed-loop transfer function:

$$T(s) = \frac{1}{\left(s^2 + 2\zeta s + 1\right)\left(\gamma s + 1\right)}.$$

This is basically a second-order part with a complex pole pair and an extra third pole at

$$s = -\frac{1}{\gamma}.$$

If that third pole is far enough left in the s-plane, it will die out fast and not affect the response much, meaning the system behaves approximately like the second order part. In the above example, the dominant second-order poles have real part of approximately $-\zeta\omega_n$, and to be able to disregard the influence of the third pole, the dominant pole rule states that

$$\left|\frac{1}{\gamma}\right| \geq |10\zeta\omega_n|$$

i.e. the third pole, should be at least 10 times farther left than the real part of the dominant pole pair. The size of $1/\gamma$ and therefore the location of the third root changes the settling time and overshoot quite a bit as shown on Figure 8.8.

γ	$\frac{1}{\gamma}$	Percent Overshoot	Settling Time*
2.25	0.444	0	9.63
1.5	0.666	3.9	6.3
0.9	1.111	12.3	8.81
0.4	2.50	18.6	8.67
0.05	20.0	20.5	8.37
0∞	20.5	8.24	

*Note: Settling time is normalized time, $\omega_n T_s$ and uses a 2% criterion.

Figure 8.8: Effect of a third pole depending on its location.

8.5 Effect of a Zero

We now consider a system with an added zero, with transfer function

$$T(s) = \frac{\left(\omega_n^2/a\right)(s+a)}{s^2 + 2\zeta\omega_n s + 1}.$$

Here, the extra zero is at $s = -a$.

A zero near the dominant poles can strongly change the transient response. In other words, if finite zeros are located relatively near the dominant complex poles, they materially affect the transient response.

A zero near the poles typically makes the step response rise faster, overshoot more, and become more “peaky’, as shown on Figure 8.9.

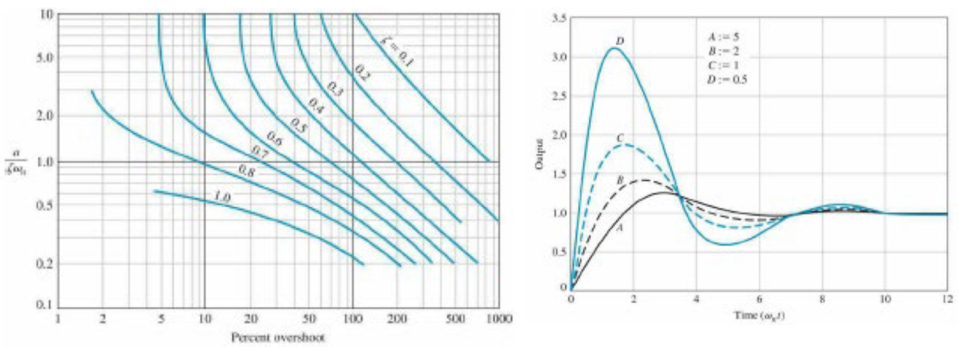


Figure 8.9: Effect of adding a zero.

Intuitively, a pole stores/slows energy, and a zero tends to add a kind of anticipatory or differentiating effect to this.

8.6 Specification Selections

We consider the system on Figure 8.10 with forward path

$$G(s) = \frac{K}{s(s+p)}$$

and unity feedback, yielding the closed-loop transfer function

$$T(s) = \frac{G(s)}{1+G(s)} = \frac{K}{s^2+ps+K}$$

which is on the same form as a standard second-order system

$$T(s) = \frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}.$$

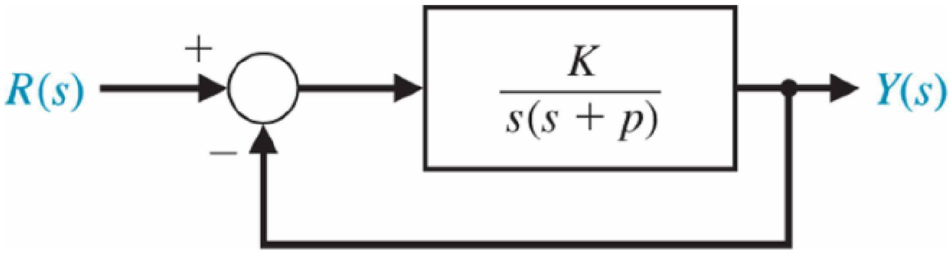


Figure 8.10: Example system.

Given this system, we want the step response to have less than 5% overshoot and a 2% settling time lower than 4 s.

For a standard second-order system, percent overshoot depends on ζ by

$$\%\text{OS} = 100e^{-\pi\zeta\frac{1}{\sqrt{1-\zeta^2}}}.$$

From Figure 8.11, an overshoot of about 4,3 % corresponds to $\zeta = 0{,}707$.

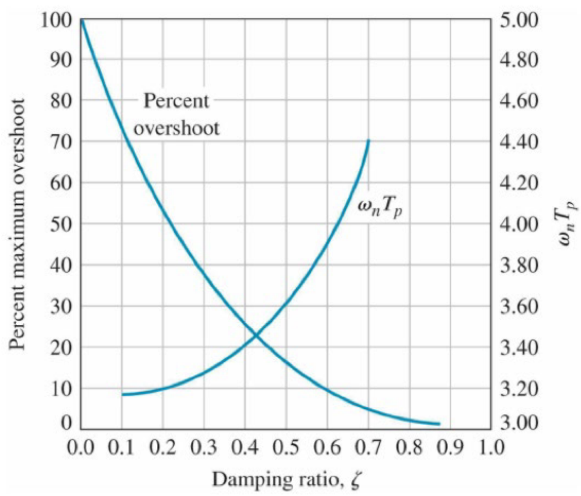


Figure 8.11: Percentage overshoot versus damping ratio.

For the 2% settling time, the standard approximation is

$$T_s \approx \frac{4}{\zeta\omega_n}.$$

We require $T_s \le 4\,\text{s}$, so

$$\frac{4}{\zeta\omega_n} \le 4 \implies \zeta\omega_n \ge 1.$$

This gives the two conditions $\zeta \ge 0{,}707$ and $\zeta\omega_n \ge 1$. For a standard underdamped second-order system, the poles are

$$s = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

i.e. ζ controls the angle of the poles and $\zeta\omega_n$ controls the real part.

On the pole plot on Figure 8.12 the 45° lines correspond to $\zeta = 0{,}707$ and the vertical line is $\zeta\omega_n = 1$, meaning real part -1. The acceptable region is therefore to the left of $\Re(s) = -1$ inside the wedge corresponding to $\zeta \ge 0{,}707$

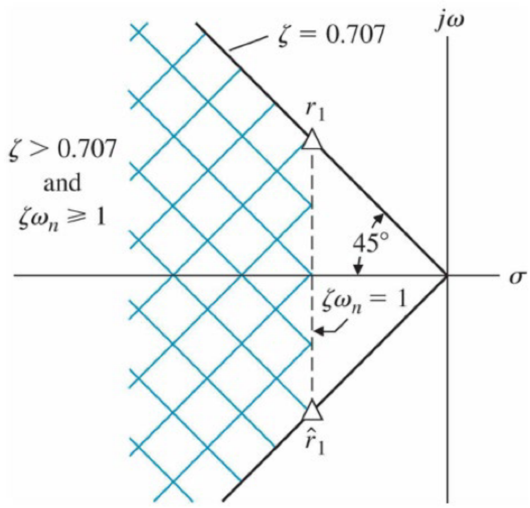


Figure 8.12: Pole plot

We choose the boundary case $\zeta = 0{,}707, \zeta\omega_n = 1$, so

$$\omega_n = \frac{1}{\zeta} = \frac{1}{0{,}707} = \sqrt{2}.$$

From earlier

$$T(s) = \frac{K}{s^2 + ps + K}.$$

Comparing this with

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}.$$

Matching coefficients gives

$$K = \omega_n^2, \qquad p = 2\zeta\omega_n.$$

Substituting, we get:

$$K = \left(\sqrt{2}\right)^2 = 2$$

and since $\zeta\omega_n = 1$, $p = 2\zeta\omega_n = 2$, so

$$K = 2, \qquad p = 2.$$

8.6.1 Estimating the Damping Ratio

The dampind ratio, ζ, can be estimated, just by looking at a step response. In general the number of visible oscillations is:

$$\text{Cycles visible} = \frac{4\beta}{2\pi\zeta} \approx \frac{0{,}55}{\zeta}.$$

So approximately

$$\zeta \approx \frac{0{,}55}{\text{cycles visible}}.$$

8.7 s-Plane Root Location and Transient Response

A transfer-function response can be decomposed into terms like

$$\frac{A_i}{s+\sigma_i} \quad \text{and} \quad \frac{B_k s + C_k}{s^2 + 2\alpha_k s + \left(\alpha_k^2 + \omega_k^2\right)}.$$

In time domain these become real exponential terms like $e^{-\sigma t}$ and oscillatory ones like $e^{-\alpha t}\sin(\omega t + \theta)$. The step-response of different root locations in the s-plane is shown on Figure 8.13.

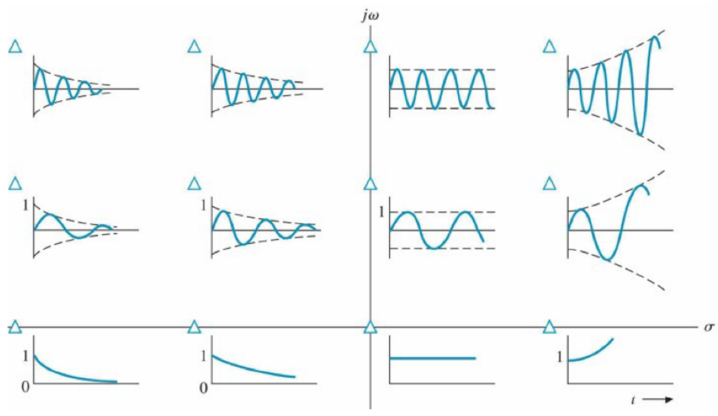


Figure 8.13: Impulse response for various root locations in s-plane

On the s-plane, poles are at $s = \sigma + j\omega$, the real part σ decides whether the response decays ($\sigma < 0$), stays constant ($\sigma = 0$), or grows ($\sigma > 0$). The imaginary part gives oscillation frequency.

8.8 Steady-state Error

We consider the system on Figure 8.14 with:

- Reference input $R(s)$
- Error $E_a(s)$
- Controller $G_C(s)$
- plant $G(s)$
- Output $Y(s)$

Because of unity feedback,

$$E(s) = R(s) - Y(s)$$

and since

$$Y(s) = G_C(s)G(s)E(s)$$

we get

$$E(s) = \frac{1}{1 + G_C(s)G(s)} R(s).$$

This is the error-transfer formula.

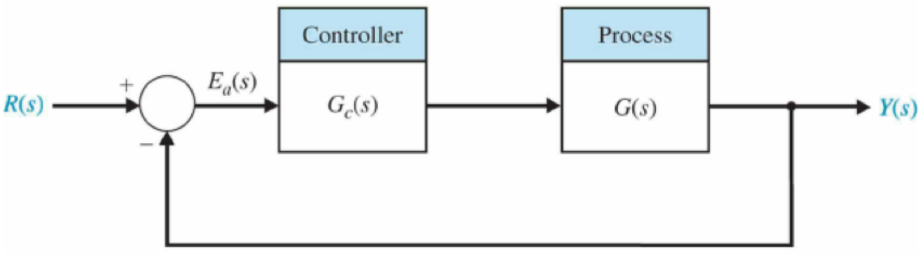


Figure 8.14: Example unity-feedback control system.

The steady-state error is the error after the transient has died out:

$$e_{ss} = \lim_{t\to\infty} e(t).$$

Using the final value theorem:

$$e_{ss} = \lim_{s\to 0} sE(s).$$

So, with the previous expression for $E(s)$

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{1 + G_c(s)G(s)} R(s). \tag{1}$$

Here, it is worth noting that $s \to 0$ means low-frequency / DC-behavior, so steady-state error depends on the system's behavior near $s = 0$.

8.8.1 Steady-State Error for Step Input

For a stem of magnitude A,

$$r(t) = A \qquad \Longrightarrow \qquad R(s) = \frac{A}{s}.$$

Plugging this into Equation (1) we get:

$$e_{ss} = \lim_{s\to 0} s \frac{1}{1 + G_C(s)G(s)} \cdot \frac{A}{s} = \frac{A}{1 + \lim_{s\to 0} G_C(s)G(s)}.$$

So, the step error depends on the DC loop gain.

The loop transfer function is written generally as:

$$G_C(s)G(s) = \frac{K\prod_{i=1}^{M}(s + z_i)}{S^N \prod_{k=1}^{Q}(s + p_k)}.$$

The S^N factor in the denominator ensures we have N poles at the origin, i.e. N *pure integrators*. This N is called the type number.

For a type 0 system ($N = 0$), $G_C(s)G(s)$ stays finite as $s \to 0$. We define the position error constant as

$$K_p = \lim_{s\to 0} G_c(s)G(s)$$

then

$$e_{ss} = \frac{A}{1+K_p}.$$

Meaning a type-0 system usually has a nonzero constant error to a step. Increasing low-frequency gain increases K_p, which reduces error, but it does not make it zero.

If the system has one or more integrators ($N \geq 1$), then $\lim_{s\to 0} G_C(s)G(s) = \infty$, so

$$e_{ss} = 0$$

for a step input. So, type-0 will have a finite step error and type-1 or higher will have zero step error.

8.8.2 Steady-State Error for Ramp Input

For a ramp with slope A,

$$r(t) = A(t) \qquad \Longrightarrow R(s) = \frac{A}{s^2}$$

then Equation (1) becomes

$$e_{ss} = \lim_{s\to 0} s \cdot \frac{1}{1+G_C(s)G(s)} \cdot \frac{A}{s^2},$$

which simplifies to a behavior controlled by $\lim_{s\to 0} sG_C(s)G(s)$. We now define the velocity error constant

$$K_v = \lim_{s\to 0} sG_C(s)G(s)$$

then for a type-1 system,

$$e_{ss} = \frac{A}{K_v}.$$

A type-0 system has no integrator, then as $sG_C(s)G(s) \to 0$, $e_{ss} = \infty$, meaning a type-0 system cannot follow a ramp forever, the error keeps growing infinitely.

For a type-1 system K_v is finite, so

$$e_{ss} = \frac{A}{K_v},$$

meaning the output follows the ramp with the same slope, but with a constant offset.

A type-2 or higher system will mean $K_v \to \infty$, so

$$e_{ss} = 0$$

meaning type-2+ can track a ramp perfectly in steady-state.

8.8.3 Steady-State Error for Accelerating/Parabolic Input

Here, the input is

$$r(t) = \frac{At^2}{2} \qquad \Longrightarrow \qquad R(s) = \frac{A}{s^3}$$

meaning the relevant term is

$$\lim_{s\to 0} s^2 G_C(s)G(s).$$

We now define the acceleration error constant $K_a = \lim_{s\to 0} s^2 G_C(s)G(s)$, and for a type-2 system we get, by insertion into Equation (1):

$$e_{ss} = \frac{A}{K_a}.$$

For a type-0 or type-1 system we do not have enough integrators to track a parabolic input so $e_{ss} = \infty$.

A type-2 system gives two integrators, so

$$e_{ss} = \frac{A}{K_a}$$

corresponding a constant finite error.

Type-3 or higher systems, gives

$$e_{ss} = 0$$

corresponding to a perfect steady-state tracking of a parabolic input.

8.9 Performance Indices

A performance index is a single number used to measure how good or bad a system response is. Usually in control, we care about how large the error is, how long the error lasts, how much overshoot there is, how quickly the system settles, etc. Instead of judging all of these from a graph, we define a mathematical quantity based on the error $e(t)$, and then try to minimize it, so a small performance index corresponds to a better response and a large performance index corresponds to a worse response.

An optimum control system is one where the system parameters are chosen so that the chosen performance index becomes as small as possible. E.g. one might tune controller gains, K_p, K_i, K_d, so that one of these errors is minimized.

All of the indices are based on the tracking error

$$e(t) = r(t) - y(t)$$

Some common performance indices are presented in the following parts.

8.9.1 Integral of the Square of the Error (ISE)

The integral of the square of the error, is defined as

$$\text{ISE} = \int_0^T e^2(t)\,\mathrm{d}t$$

meaning you square the error and add it up over time. This tends to favor responses that reduce large errors quickly, but as small errors are not weighted as much oscillation is not impeded much.

8.9.2 Integral of the Absolute Error (IAE)

The integral of the absolute error, is defined as

$$\text{IAE} = \int_0^T |e(t)|\,\mathrm{d}t$$

meaning you take the absolute value of the error and add it over time. This does not exaggerate large errors as much as ISE and generally treats errors more "linearly".

8.9.3 Integral of Time Multiplied by Absolute Error (ITAE)

The integral of the time multiplied by absolute error, is defined as

$$\text{ITAE} = \int_0^T t\,|e(t)|\,\mathrm{d}t$$

meaning the absolute error is now multiplied by the time t before being integrated. I.e. errors at the beginning count less and errors later count more. This generally helps to reduce settling time and long-lasting oscillation.

8.9.4 Integral of Time multiplied by the Squared Error (ITSE)

The integral of time multiplied bu the squared error is defined as

$$\text{ITSE} = \int_0^T t e^2(t)\,\mathrm{d}t.$$

This combines the ideas of squaring the error to emphasise large errors, and multiplying by the time to emphasise late errors.

8.9.5 The factor T

The factor T in all the above integrals is usually chosen as the settling time T_s. This means the integral is often evaluated only over a practical time window. In theory, people sometimes also use

$$\int_0^\infty \ldots\,\mathrm{d}t$$

if the response dies out and the integral remains finite.

8.9.6 Simplification of Linear Systems

We have two main methods if we want to approximate a high-order transfer function as a lower-order one that keeps the most important behavior.

The first method is deleting an insignificant pole. E.g.

$$G(s) = \frac{K}{s(s+2)(s+30)}$$

with poles at $s = 0, -2, -30$. Here the pole at -30 corresponds to a very fast mode that dies out much faster than the other two, so for the slower response, we approximate

$$s + 30 \approx 30$$

which gives

$$G(s) \approx \frac{K}{30s(s+2)} = \frac{K/30}{s(s+2)}.$$

This is called a *dominant-pole approximation.*

The second method is matching the frequency response. Instead of just deleting a pole, you can instead build a lower-order model whose frequency response is close to the original one. Say for a high-order system

$$G_H(s) = K\frac{a_m s^m + a_{m-1}s^{m-1} + \ldots + a_1 s + 1}{b_n s^n + b_{n-1}s^{n-1} + \ldots + b_1 s + 1}$$

we can create a lower-order approximation

$$G_L(s) = K\frac{c_p s^p + \ldots + c_1 s + 1}{d_q s^q + \ldots + d_1 s + 1}$$

by choosing the c_i and d_i so that $G_L(s)$ behaves as much like $G_H(s)$ as possible in frequency. In general, to match frequency response, the ratio

$$\frac{G_H(s)}{G_L(s)} = \frac{M(s)}{\Delta(s)} \to 1$$

I.e. if $G_L(s)$ is a good approximation, then the ratio $G_H(s)/G_L(s)$ should be close to 1. So the goal is to make the numerator and denominator of that ratio behave similarly. We can match using,

$$M_{2q} = \Delta_{2q}$$

for $q = 0, 1, 2, \dots$ I.e. you compare certain coefficients derived from $M(s)$ and $\Delta(s)$ and force them to be equal.

Example: Frequency Response Matching

We seek to approximate the system

$$G_H(s) = \frac{6}{s^3 + 6s^2 + 11s + 6}$$

by a second-order model

$$G_L(s) = \frac{1}{1 + d_1 s + d_2 s^2}.$$

We start by dividing the denominator by 6

$$G_H(s) = \frac{1}{1 + \frac{11}{6}s + s^2 + \frac{1}{6}s^3}.$$

Now, we define

$$\Delta(s) = 1 + \frac{11}{6}s + s^2 + \frac{1}{6}s^3$$

and for the approximate model

$$M(s) = 1 + d_1 s + d_2 s^2$$

because

$$\frac{G_H(s)}{G_L(s)} = \frac{1/\Delta(s)}{1/M(s)} = \frac{M(s)}{\Delta(s)}.$$

Now we compute the derivatives at $s = 0$. For $\Delta(s)$:

$$\Delta^{(0)}(0) = 1, \quad \Delta^{(1)}(0) = \frac{11}{6}, \quad \Delta^{(2)}(0) = 2, \quad \Delta^{(3)}(0) = 1$$

and for $M(s)$:

$$M^{(0)}(0) = 1, \quad M^{(1)}(0) = d_1, \quad M^{(2)}(0) = 2d_2, \quad M^{(3)}(0) = 0.$$

Now, we match the coefficients, by first computing

$$M_2 = d_1^2 - 2d_2$$

and

$$\Delta_2 = \frac{49}{36}.$$

So matching gives

$$d_1^2 - 2d_2 = \frac{49}{36}.$$

The next matching condition gives

$$M_4 = \Delta_4$$

which becomes

$$d_2^2 = \frac{7}{18} \implies d_2 \approx 0{,}624 \implies d_1 1{,}62.$$

So

$$G_L(s) \approx \frac{1}{1+1{,}62s+0{,}624s^2}.$$

Lecture 5: Stability of Linear Feedback Systems 7. April 2026

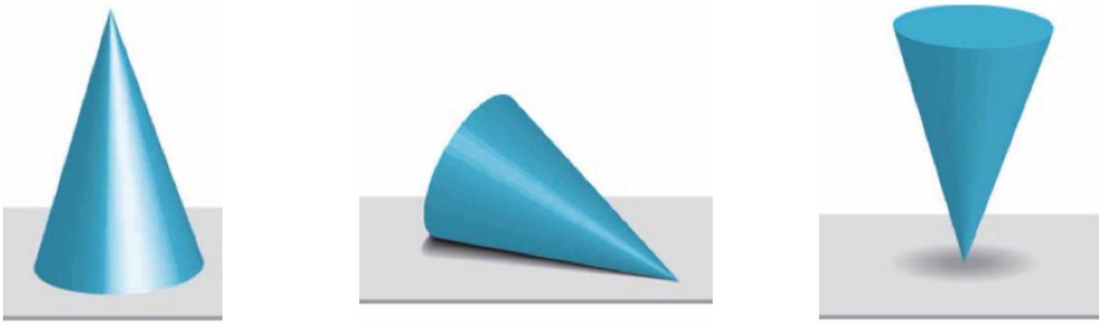
9 Stability of Linear Feedback Systems

9.1 Stability Definition

Stability is often the most important concept in control systems. A system might have a nice speed, accuracy, and disturbance rejection, but none of that matters if the system is unstable, because the output can then grow uncontrollably.

Definition 9: Stable System

A stable system is any system that gives a bounded output for any bounded input. This is also called *BIBO*-stability (Bounded-Input, Bounded-Output). Stability is illustrated by the different cones on Figure 9.1.



(a) Stable (b) Neutral (c) Unstable

Figure 9.1: a cone in different configurations corresponding to a stable, neutral, and unstable system.

On Figure 9.1, a cone is shown in three different configurations. Figure 9.1a shows a cone standing on its base. If you disturb it a little, it stays in a reasonable state (on its base). Here, the system does not "run away" due to a small input. On Figure 9.1b a cone lying on its side is shown. Here, if you move it, it stays where you put it rather than returning or diverging (ideally). This corresponds to marginal/neutral stability. Figure 9.1c represents an unstable system by a cone balanced on its tip. Here, any small disturbance makes the cone fall away.

9.2 Stability Analysis with s-plane Approach

On Figure 8.4 a correlation between different pole locations and the corresponding time responses was shown. This is summarized on Figure 9.2.

Here, we see that a system is stable if all poles have negative real parts, and the system is unstable if any pole has positive real part. Marginally stable systems are the special case, where poles are on the imaginary axis (real part equal to zero) and they are simple poles, while all other poles have negative real parts. This happens, as a pole contributes a term like

$$e^{st} = e^{(\sigma+j\omega)t} = e^{\sigma t}e^{j\omega t}$$

so, if $\sigma < 0$, $e^{\sigma t}$ decays with time and vice versa. For $\sigma = 0$, $e^{\sigma t} = 1$ so the response will have sustained oscillation with no decay or growth.

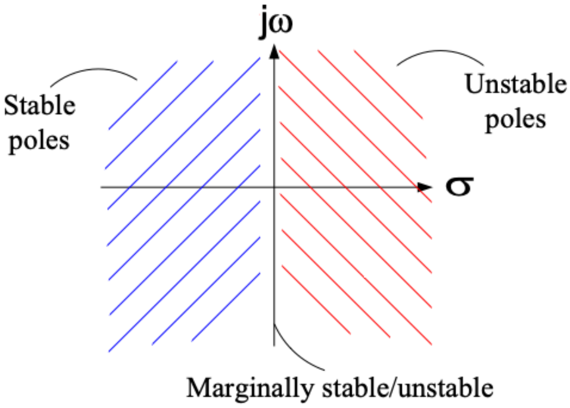


Figure 9.2: Stability of poles depending on their location in the s-plane

9.2.1 Relative Stability of Feedback Control Systems

Stability is rarely binary. In general, we work with two different ideas: Absolute stability and relative stability.

Absolute stability is a simply yes/no question: "Is the system stable or not?". If all poles are in the left half-plane, the system is absolutely stable.

Relative stability instead concerns itself with how stable a system is, how close the poles are to instability, and how oscillatory the response is.

These two concepts are compared on Figure 9.3.

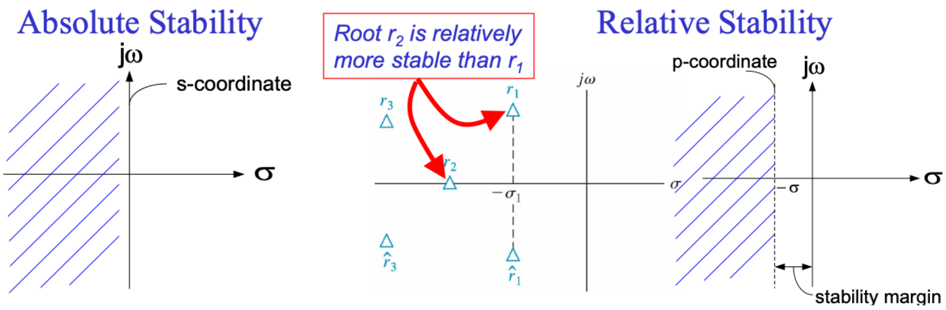


Figure 9.3: Comparison of different ideas of stability.

On Figure 9.3, it is stated that r_2 is relatively more stable than r_1. This is true, as r_2 is farther left in the s-plane, so its real part is more negative, meaning it is associated with a faster response decay.

The right figure on Figure 9.3 shows a vertical line left of the imaginary axis. The distance from that line to the imaginary axis is called the stability margin. Here, we do not just want poles barely in the left half-plane. We want them *comfortably* to the left, so the system remains well-behaved even with modelling errors or parameter variations.

For complex poles, relative stability is also described by the damping ratio. Higher damping ratios leads to less oscillation and better settling corresponding to better relative stability.

9.3 Routh-Hurwitz Stability Criterion

For a closed-loop LTI system, stability is determined by the roots of the characteristic polynomial

$$q(s) = a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0$$

these roots are the closed loop poles.

The Routh-Hurwitz method is a way to determine whether the roots of the characteristic equation have negative real parts just by looking at the coefficients of said equation. Here, we use the Hurwitz's necessary condition stating that if $a_n > 0$, then for stability all coefficients must be positive. This is only a necessary and not a sufficient condition. I.e. a stable system will necessarily have positive coefficients to the characteristic equation, but having positive coefficients does not mean a system is necessarily stable. This is exactly why Routh-Hurwitz is needed as it gives a stronger test.

9.3.1 Routh Array

To use the Routh-Hurwitz method, we start by building a so-called Routh array from the polynomial coefficients. For

$$a_n s^n + a_{n-1} s^{n-1} + a_{n-2} s^{n-2} + \ldots + a_0.$$

The Routh array is defined as:

$$\begin{array}{c|cccc} s^n & a_n & a_{n-2} & a_{n-4} \\ s^{n-1} & a_{n-1} & a_{n-3} & a_{n-5} \\ s^{n-2} & b_{n-1} & b_{n-3} & b_{n-5} \\ s^{n-3} & c_{n-1} & c_{n-3} & n_{n-5} \\ \vdots & \vdots & \vdots & \vdots \\ s^0 & h_{n-1} & & \end{array}$$

where

$$\begin{aligned} b_{n-1} &= \frac{a_{n-1}a_{n-2} - a_n a_{n-3}}{a_{n-1}} = -\frac{1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-2} \\ a_{n-1} & a_{n-3} \end{vmatrix} \\ b_{n-3} &= -\frac{1}{a_{n-1}} \begin{vmatrix} a_n & a_{n-4} \\ a_{n-1} & a_{n-5} \end{vmatrix} \\ c_{n-1} &= -\frac{1}{b_{n-1}} \begin{vmatrix} a_{n-1} & a_{n-3} \\ b_{n-1} & b_{n-3} \end{vmatrix}. \end{aligned}$$

I.e. the first two rows are arranged as a_n, a_{n-2}, a_{n-4} and $a_{n-1,a_{n-3}}, a_{n-5}$, respectively and the next rows are then computed from the previous two.

The important result here, is then that the number of roots of $q(s)$ with positive real parts equals the number of sign changes in the first column of the Routh array.

Example: Use of the Routh-Hurwitz Stability Criterion

We are given a characteristic polynomial in factored form as

$$q(s) = \left(s - 1 + j\sqrt{7}\right)\left(s - 1 - j\sqrt{7}\right)(s+3).$$

From this we see that the roots are $s = 1 + \pm j\sqrt{7}, -3$ so the first two have real part $1 > 0$, so two roots are in the right-half plane, meaning the system is unstable. Expanding the polynomial we get:

$$q(s) = s^3 + s^2 + 2s + 24.$$

Here, all coefficients are positive, yet the system is still unstable, which directly proves the statement that the Hurwitz's necessary condition is not sufficient but only necessary.

For a cubic

$$a_3 s^3 + a_2 s^2 + a_1 s + a_0$$

the Routh array is

$$\begin{array}{c|cc} s^3 & a_3 & a_1 \\ s^2 & a_2 & a_0 \\ s^1 & b_1 & 0 \\ s^0 & c_1 & 0 \end{array}.$$

Here, $a_3 = 1, a_2 = 1, a_1 = 2, a_0 = 24$, so the first two rows are:

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 1 & 24 \end{array}.$$

Now we compute

$$b_1 = \frac{a_2 a_1 - a_3 a_0}{a_2} = \frac{1 \cdot 2 - 1 \cdot 24}{1} = -22$$

so the last row becomes

$$c_1 = a_0 = 24$$

meaning the array is

$$\begin{array}{c|cc} s^3 & 1 & 2 \\ s^2 & 1 & 24 \\ s^1 & -22 & 0 \\ s^0 & 24 & 0 \end{array}.$$

Here, there are two sign changes, so we have two 2 roots with positive real part meaning the system is unstable, which matches the expected result from the introduction.

Lecture 6: Root Locus Method, I 9. April 2026

10 The Root Locus Method

10.1 Introduction and Concepts

For the unity-feedback system,

$$\frac{Y(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)}$$

the closed-loop poles are the roots of the denominator, so they satisfy

$$1 + KG(s) = 0.$$

This equation is the characteristic equation.

So root locus means that we pick a value of K, solve $1 + KG(s) = 0$, plot the resulting closed-loop poles in the s-plane and then vary K from 0 to ∞. The curve traced by these poles is the root locus.

We can rewrite the above characteristic equation as

$$KG(s) = -1$$

which is useful because -1 has magnitude 1 and angle 180° modulo 360°. This means we can split the condition into two parts:

$$|KG(s)| = 1$$
$$\angle KG(s) = 180^\circ + k360^\circ.$$

The key idea then is that changing G changes the magnitude, but not the angle if $K > 0$. So the angle condition tells where the root locus can exist and the magnitude condition tells which value of K places a pole at that point.

Example: Example system

We consider the example system on Figure 10.1 with open-loop transfer function

$$G(s) = \frac{1}{s(s+2)}$$

and gain K, so

$$KG(s) = \frac{K}{s(s+2)}.$$

Which means that the open-loop poles are at $s = 0$ and $s = -2$. There are no zeros. The two roots are where the root-locus branches start when $K = 0$.
Using

$$1 + KG(s) = 0 = 1 + \frac{K}{s(s+2)}$$

we multiply by $s(s+2)$ to get

$$s(s+2) + K = 0$$

so

$$s^2 + 2s + K = 0$$

which is the closed-loop characteristic equation. We can compare this to the standard second-order form

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0.$$

Matching coefficients gives:

$$2\zeta\omega_n = 2 \quad \text{and} \quad \omega_n^2 = K.$$

So

$$\zeta\omega_n = 1, \qquad \omega_n = \sqrt{K}.$$

Which means

$$\zeta = \frac{1}{\sqrt{K}}.$$

So as K increases ω_n also increases, but ζ decreases.
Solving

$$s^2 + 2s + K = 0$$

gives

$$s_{1,2} = -1 \pm \sqrt{1-K}.$$

When $0 < K < 1$, $\sqrt{1-K}$ is real, so both poles are real. One will move rightward from -2 and one will move leftward from 0 approaching each other along the real axis.
When $K = 1$ $s = -1$ so both poles meet at -1 which is the breakaway point.
When $K > 1$, $\sqrt{1-K} = j\sqrt{K-1}$, so the poles are $s = -1 \pm j\sqrt{K-1}$, from where they move vertically upward

and downward, with constant real part -1.
Further, for this system

$$KG(s) = \frac{K}{s(s+2)}.$$

At a candidate point $s = s_1$, the angle condition is

$$\angle \frac{K}{s(s+2)} = -\angle s - \angle(s+2) = 180^\circ \mod 360^\circ.$$

Because there are no zeros, the total angle comes only from the poles. On the vertical line through -1, the geometry is symmetric, with one pole contributing angle θ and one contributing angle $180^\circ - \theta$, which means the total denominator angle is 180°, which means the angle condition is satisfied.
After we know a point is on the root locus from the angle condition, the magnitude condition gives the corresponding gain

$$\left|\frac{K}{s(s+2)}\right| = 1.$$

so

$$K = |s|\,|s+2|.$$

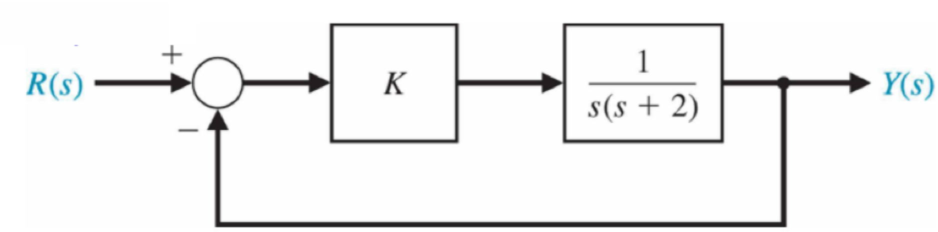


Figure 10.1: Example 2nd order control system

In general, if the loop transfer function is

$$F(s) = K\frac{\prod_{i=1}^{M}(s+z_i)}{\prod_{k=1}^{n}(s+p_k)}$$

then the characteristic equation is

$$1 + F(s) = 0$$

and the root locus conditions become

$$K\left|\frac{\prod(s+z_i)}{\prod(s+p_k)}\right| = 1$$
$$\sum\angle(s+z_i) - \sum\angle(s+p_k) = 180^\circ + k360^\circ.$$

So zeros add angle, whilst poles subtract angle.

10.2 The Root Locus Procedure

The following part will outline the general procedure for drawing a root-locus sketch.

10.2.1 Prepare the Root Locus Sketch

We should start by rewriting the characteristic equation so the gain appears as a multiplier

$$1 + KP(s) = 0.$$

Then factor $P(s)$ into poles and zeros.

The branches will always start at poles and end at zeros as

$$1 + K\frac{\prod(s+z_i)}{\prod(s+p_k)} = 0$$

then, when $K = 0$ the equation reduces to the denominator being zero, so the roots are the open-loop poles, and when $K \to \infty$, the numerator dominates, so the roots approach the open-loop zeros. So the root locus begins at poles and ends at zeros. If there are fewer finite zeros than poles, then the extra branches go to zeros at infinity. Also, because the coefficients are real, the locus is symmetri about the real axis.

Example: Preparing the Locus sketch

We are given an example system

$$1 + \frac{K}{s^4 + 12s^3 + 64s^2 + 128s} = 0.$$

And we want to apply the above procedure to prepare a root locus sketch.

We factor the denominator:

$$\begin{aligned} s^4 + 12s^3 + 64s^2 + 128s &= s(s+4)\left(s^2 + 8s + 32\right) \\ &= s\,(s+4))\left(s+4+4j\right)\left(s+4-4j\right). \end{aligned}$$

So the open loop poles are $s = 0, -4, -4+4j, -4-4j$, and there are no finite zeros. So the number of poles $n = 4$, the number of zeros $M = 0$, and the number of branches is 4.

10.2.2 Find the real-axis part of the locus

We have a general rule stating that a point on the real axis is on the root locus if the number of poles and zeros to its right is odd.

For the above example, we have 0 singularities to the right of 0, so anything to the right of 0 is not on the locus. Between −4 and 0, we have 1 singularity on the right so this part of the real axis is on the locus. To the left of −4, we have 2 singularities to the right so this part of the real axis is not on the locus. This means the only real-axis segment on the root locus is

$$-4 < s < 0.$$

10.2.3 Asymptotes to Infinity

Since there are more poles than zeros, some branches must go to infinity. The number of asymptotes is

$$n - M = 4 - 0 = 4.$$

The asymptote angles are

$$\phi_A = \frac{(2k+1)180°}{n-M}, \qquad k = 0,1,2,3.$$

So here:

$$\phi_A = 45°, 135°, 225°, 315°.$$

The asymptote centroid is found as

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n-M}.$$

Here, the poles sum to -12 and the zeros sum to 0, so

$$\sigma_A = \frac{-12}{4} = -3.$$

So the four asymptotes all cross the real axis at $s = -3$, with angles $45°, 135°, 225°, 315°$.

10.2.4 Crossing Between Root Locus and the Imaginary Axis

Here, we use the Routh-Hurwitz criterion. We start by rewriting the characteristic equation as a polynomial:

$$s^4 + 12s^3 + 64s^2 + 128s + K = 0.$$

So our Routh array is

$$\begin{array}{c|ccc} s^4 & 1 & 64 & K \\ s^3 & 12 & 128 & 0 \\ s^2 & \frac{12\cdot 64 - 1\cdot 128}{12} & K & 0 \\ s^1 & \frac{\frac{640}{12}\cdot 128 - 12K}{\frac{640}{12}} & 0 & 0 \\ s^0 & K & & \end{array}.$$

The s^2 row first term is $(768-128)/12 = 53{,}3$ and the s^1 ros first term becomes zero at the crossing

$$\frac{53{,}3\cdot 128 - 12K}{53{,}3} = 0$$

which gives $K \approx 569$.

Now we use

$$53{,}3s^2 + 569 = 0$$

so $s^2 = -10{,}7 \implies s \pm j3{,}27$.

So the root locus crosses the imaginary axis at $s \pm j3{,}27$ when $K = 569$.

10.2.5 Breakaway Points

A breakaway point is where two branches meet on the real axis and then leave it into the complex plane.

Since the real-axis locus is only between -4 and 0, the poles at 0 and -4 move toward each other along that segment, meet, and then break away.

From

$$1 + \frac{K}{D(s)} = 0 \implies D(s) + K = 0 \implies K = -D(s)$$

where

$$D(s) = s(s+4)(s+4+4j)(s+4-4j).$$

The breakaway points satisfy

$$\frac{\mathrm{d}K}{\mathrm{d}s} = 0$$

which is equivalent to differentiating $D(s)$. The real solution on the locus is $s \approx -1{,}58$, so the branches from 0 and -4 meet around $s = -1{,}58$ and leave the real axis there.

10.2.6 Angle of departure from the complex poles

Because there are complex poles at

$$-4 \pm 4j$$

the branches do not leave those poles in arbitrary directions. They must satisfy the angle condition

$$\angle P(s) = (2k+1)180°.$$

For the upper pole $-4+4j$, the angles contributed by the other poles are used to determine the departure angle. We can find one equivalent angle as 225°, and because angles are modulo 360°, that is the same as $-135°$. So the key-idea is that the departure direction is forced by the angle condition, and that the two complex-pole branches leave symmetrically.

10.2.7 Complete the Sketch

Now all the pieces are known. We have poles at 0, -4, and $-4 \pm 4j$, no finite zeros and the only part of the real axis on the locus is $(-4, 0)$. The asymptotes are through -3 at 45° , 135° , 225° , and 315°. The breakaway is at $-1{,}58$ and the imaginary crossing is at $\pm j3{,}27$ when $K = 569$.

10.3 Parameter Design by the Root Locus Method

Suppose a system has a characteristic equation on the form

$$a_n s^n + a_{n-1} s^{n-1} + \ldots + a_1 s + a_0 = 0$$

and we then want to know what happens when one coefficient, say a_1, changes.

We can rearrange the above into the root-locus form:

$$1 + \frac{a_1 s}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_2 s^2 + a_0} = 0$$

so now we can interpret it as a gain-like parameter $K = a_1$, and an open-loop part:

$$P(s) = \frac{s}{a_n s^n + a_{n-1} s^{n-1} + \ldots + a_2 s^2 + a_0}.$$

So even though a_1 is not the loop gain, it can still be treated as the parameter that “scales” the transfer function. Then the root locus tells us where the closed-loop poles go as a_1 changes, whether stability improves or worsens, or how damping and speed change.

10.3.1 Varying Multiple Parameters

If we instead have a characteristic equation like

$$s^3 + s^2 + \beta s + \alpha = 0$$

and we want to adjust both the α and β parameters, then we cannot directly make an ordinary root locus that simultaneously sweeps both α and β as this method can only vary one scalar parameter at a time. Instead we repeat the procedure twice.

We start by studying α by setting $\beta = 0$. Then the characteristic equation becomes

$$s^3 + s^2 + \alpha = 0 \implies 1 + \frac{\alpha}{s^3 + s^2} = 0.$$

Now this is a root locus with parameter $K_2 = \alpha$ and tranfer function:

$$P_2(s) = \frac{1}{s^3 + s^2}.$$

We then use this root locus to choose a useful value of α based on the pole locations we want.

Then we fix α at the chosen value and study β, by rewriting the equation as

$$1 + \frac{\beta s}{s^3 + s^2 + \alpha} = 0$$

so now the parameter is $K_1 = \beta$ and the transfer function is

$$P_1(s) = \frac{s}{s^3 + s^2 + \alpha}.$$

This, in turn, lets us see how the poles move as β changes.

Lecture 7: Root Locus Method, II

14. April 2026

11 PID Controllers

A PID-controller (Proportional-Integral-Derivative) takes the error signal

$$e(t) = r(t) - y(t)$$

and turns it into a control input $u(t)$ for a plant/process. A block diagram for how this process might look is shown on Figure 11.1.

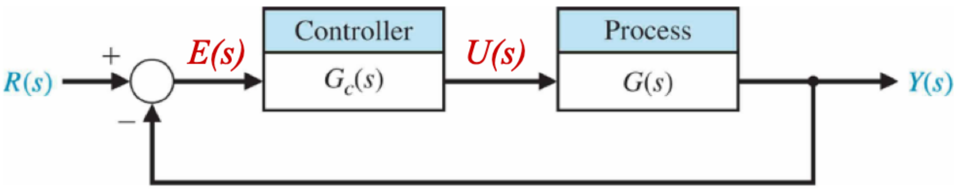


Figure 11.1: Block diagram of a simple process with a controller and feedback.

The controller is

$$u(t) = K_P e(t) + K_I \int e(t)\,\mathrm{d}t + K_D \frac{\mathrm{d}e(t)}{\mathrm{d}t}$$

or in the s-domain

$$G_c(s) = K_P + \frac{K_I}{s} + K_D s. \tag{2}$$

Here, the three terms K_P, K_I, and K_D each correspond to different elements as shown in Table 1. In this way, the

Table 1: Meaning of the different terms in a PID-controller.

Term	Meaning	Effect
$K_P e(t)$	Proportional	Reacts to present error
$K_I \int e(t)\,\mathrm{d}t$	Integral	Reacts to past error; Removes steady-state error
$K_D \frac{\mathrm{d}e(t)}{\mathrm{d}t}$	Derivative	Reacts to change in error; Adds damping

output of a PID-controller is based on both the present error, the past accumulated error, and the rate in change of the error.

11.1 Forms of PID Controllers

The ideal derivative term in Equation (2) is written as $K_D s$, however, a pure differentiator is usually not implemented directly, as it amplifies high frequency noise. In practice, the derivative part is often filtered as

$$G_d(s) = \frac{K_D s}{\tau_d s + 1}$$

which is approximately equal to $K_D s$ when τ_d is very small compared with the process time constants.

It is also possible to leave out the derivative part entirely leaving you with a PI-controller with transfer function:

$$G_c(s) = K_P + \frac{K_I}{s}.$$

Or a PD-controller with transfer function:

$$G_c(s) = K_P + K_D s.$$

A PI-controller is common when steady-state error matters, whereas a PD-controller is more common when damping/overshoot reduction is important but the integral action is not as useful.

11.1.1 PID as a Cascade of PI and PD

A PID-controller can be interpreted as a PI controller followed by a PD controller:

$$G_c(s) = G_{PI}(s) G_{PD}(s)$$

where

$$\begin{aligned} G_{PI}(s) &= \hat{K}_P + \frac{\hat{K}_I}{s} \\ G_{PD}(s) &= \overline{K}_P + \overline{K}_D s. \end{aligned}$$

Multiplying these (as we do when combining cascading blocks) gives

$$G_C(s) = \left(\hat{K}_P + \frac{\hat{K}_I}{s}\right)\left(\overline{K}_P + \overline{K}_D s\right) = \hat{K}_P \overline{K}_P + \hat{K}_P \overline{K}_D s + \frac{\hat{K}_I \overline{K}_P}{s} + \hat{K}_I \overline{K}_D.$$

We see that this is a PID-controller with

$$\begin{aligned} K_P &= \overline{K}_P \hat{K}_P + \hat{K}_I \overline{K}_D \\ K_D &= \hat{K}_P \overline{K}_D \\ K_I &= \hat{K}_I \overline{K}_P. \end{aligned}$$

11.1.2 Pole-zero interpretation of a PID controller

We use the general form of the transfer function for a PID controller, Equation (2), and put everything over a common denominator

$$G_c(s) = \frac{K_D s^2 + K_P s + K_I}{s}.$$

Factoring out K_D we get:

$$G_C(s) = \frac{K_D\left(s^2 + \frac{K_P}{K_D} s + \frac{K_I}{K_D}\right)}{s}.$$

Then the quadratic part of the numerator can be written as

$$s^2 + \alpha s + b, \qquad \alpha = \frac{K_P}{K_D}, \quad b = \frac{K_I}{K_D},$$

which can be factored to $(s+z_1)(s+z_2)$. So:

$$G_c(s) = \frac{K_D(s+z_1)(s+z_2)}{s}.$$

This means a PID-controller contributes one pole at $s = 0$ and two zeros determined by K_P, K_I, and K_D. The pole at the origin comes from the integral term, K_I/s. The two zeros come from the numerator:

$$K_D s^2 + K_p s + K_I.$$

By choosing K_P, K_I, and K_D, you change where these zeros are located. In root locus design this is powerful as these zeros can then be used to pull the root locus toward the desired regions of the s-plane.

11.2 PID Tuning

As mentioned in Section 11.1.2, the values of K_P. K_I, and K_D can be tuned to change the root locus. This can generally be done in two different ways, namely via manual trial-and-error tuning with minimal analytic analysis using step responses and via Ziegler-Nichols tuning, which is a more analytic tuning method based on closed- and open-loop response to a step input.

11.2.1 Manual PID Tuning

Manual tuning starts by simplifying the controller as much as possible by setting $K_I = K_D = 0$, so initially we only have

$$G_c(s) = K_P.$$

Which means, first we must tune a proportional controller. We then increase K_P until the closed-loop system is just about unstable, before we reduce K_P until the response has roughly quarter-amplitude decay. Then we increase K_I and K_D manually to improve the step response. Here, "quarter-amplitude decay" means that oscillations die out such that each overshoot peak os about one quarter of the previous overshoot peak. This is a classic tuning target as it is neither too sluggish nor too oscillatory for most applications.

Effect of each PID gain Table 2 summarizes the usual qualitative effects of increasing each PID gain. The

Table 2: Qualitative effect of increasing each PID gain

Increasing	Overshoot	Settling Time	Steady-state error
K_P	Increases	Small/minimal effect	Decreases
K_I	Increases	Increases	Removes steady-state error
K_D	Decreases	Decreases	Little/no direct effect

intuition here is that K_P pushes harder when the error is large. This usually makes the system respond faster, but it can also lead to more overshoot. K_I keeps accumulating error over time, which is excellent for removing steady-state error, but it often makes the system more oscillatory. K_D reacts to the rate of change of the error acting like a damper. It tends to reduce overshoot and make oscillations die out faster.

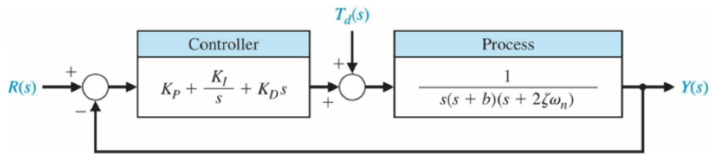


Figure 11.2: Example system used for the manual tuning example.

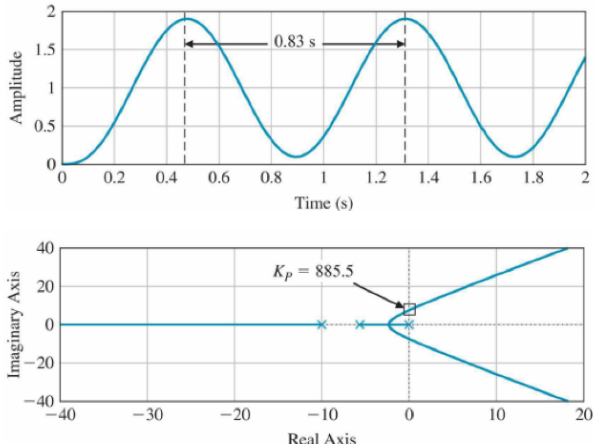


Figure 11.3: Root locus of P-controller at marginal stability.

Example: Manual Tuning of a PID-controller

We consider the system on Figure 11.2. Here, the plant has the transfer function

$$G(s) = \frac{1}{s(s+b)(s+2\xi\omega_n}$$

with $b = 10, \xi = 0{,}707, \omega_n = 4$. Hence, the process is approximately

$$G(s) = \frac{1}{s(s+10)(s+5{,}66)}.$$

So the open-loop system has poles at $s = 0, s = -10, s = -5{,}66$. The pole at the origin means the plant already contains an integrator

---

We start by setting $K_I = K_D = 0$, leaving us with

$$G_c(s) = K_P.$$

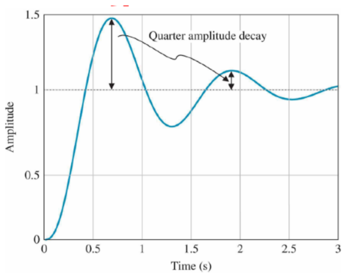
This gives the closed-loop characteristic equation:

$$1 + K_P \frac{1}{s(s+10)(s+5{,}66)} = 0$$

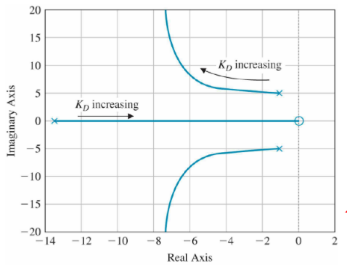
or

$$s(s+10)(s+5{,}66) + K_P = 0.$$

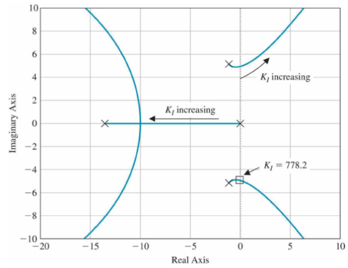
When $K_P = 886$, the closed-loop poles are exactly at $s = \pm 7{,}5j$, which is exactly on the imaginary axis. This is shown on Figure 11.3.
Here, the response oscillates forever without growing or decaying – This is also known as marginal stability. The oscillation period is approximately $T = 2\pi/7{,}5 = 0{,}84\,\mathrm{s}$ Since $K_P = 886$ gives sustained oscillation, this



(a) Quarter-Amplitude decay case.



(b) Varying K_D in the PD-controller.



(c) Varying K_I in the PI-controller.

is too aggressive. We therefore reduce K_P to about 370 to obtain quarter-amplitude decay. This is shown on Figure 11.4a.
We now fix $K_P = 370, K_I = 0$ and vary K_D, which means the controller is now a PD-controller:

$$G_c(s) = K_P + K_D s.$$

The root locus for this is shown on Figure 11.4b, where we see that increasing K_D moves the complex poles left, corresponding to faster decay.
If we now fix $K_P = 370, K_D = 0$ and instead vary K_I the controller is instead a PI-controller:

$$G_c(s) = K_P + \frac{K_I}{s}.$$

On Figure 11.4c, we see that increasing K_I moves the complex poles to the right. At $K_I = 778$, the system becomes marginally stable with poles at $s = \pm 4{,}86j$.
Finally, we choose $K_P = 370, K_I = 100, K_D = 75$ giving the PID-controller

$$G_c(s) = 370 + \frac{100}{s} + 60s = \frac{60s^2 + 370s + 100}{s}.$$

This has the step response depicted on Figure 11.5 with overshoot of approximately 8,78 % and settling time of approximately 3,24 s.

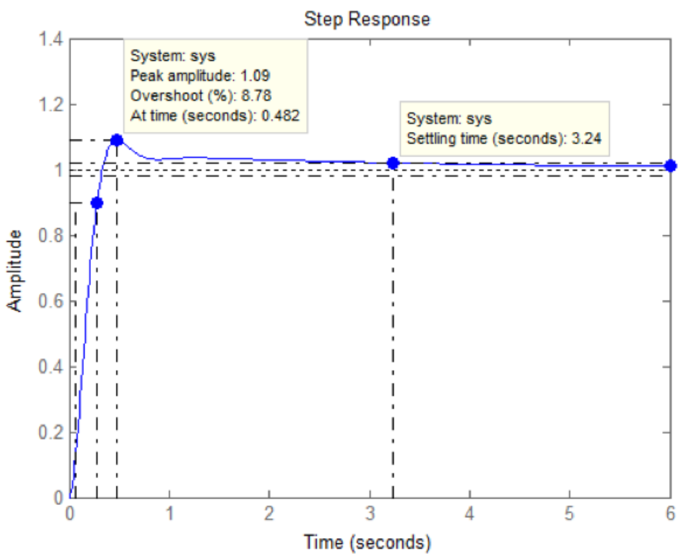


Figure 11.5: Manually tuned PID-controller.

11.2.2 Closed-loop Ziegler-Nichols PID Tuning

Ziegler-Nichols tuning is a rule-based way to get initial PID gains without needing a perfect mathematical model of the process. We generally distinguish between closed-loop and open-loop Ziegler Nichols tuning. Closed-loop Ziegler-Nichols tuning uses the closed-loop system and finds the ultimate gain and ultimate period, whereas open-loop Ziegler-Nichols tuning uses the open-loop step response, also called the reaction curve. Here, we focus on the closed-loop Ziegler-Nichols method.

The method starts like manual tuning with $K_I = K_D = 0$, so the controller is initially only proportional:

$$G_c(s) = K_P.$$

Then K_P is increased, until the closed-loop system is on the boundary of instability, i.e. marginal stability. At this point we measure two things, K_U which is the ultimate gain (the value of K_P that gives sustained oscillations), and the ultimate period T_U (the period of the sustained oscillations).

The closed-loop Ziegler-Nichols rules are shown on Table 3.

Table 3: Closed-loop Ziegler-Nichols rules

Controller	K_P	K_I	K_D
P	$0{,}5K_U$	–	–
PI	$0{,}45K_U$	$\frac{0{,}54K_U}{T_U}$	–
PID	$0{,}6K_U$	$\frac{1{,}2K_U}{T_U}$	$\frac{0{,}6K_U T_U}{8}$

Example: Ziegler-Nichols Tuning of a PID-controller

We use the same process as before:

$$G(s) = \frac{1}{s(s+b)(s+2\xi\omega_n)} = \frac{1}{s(s+10)(s+5{,}66)}.$$

With $K_I = K_D = 0$, the closed-loop characteristic equation is

$$1 + K_P \frac{1}{s(s+10)(s+5{,}66)} = 0.$$

When $K_P = 886$ the system is at marginal stability and the closed-loop poles are at $s = \pm 7{,}5j$ as shown on Figure 11.3. Therefore $K_U = 886$ and the period of oscillation here is $T_U = 0{,}84\,\text{s}$.
Using the PID-row of Table 3, we see that:

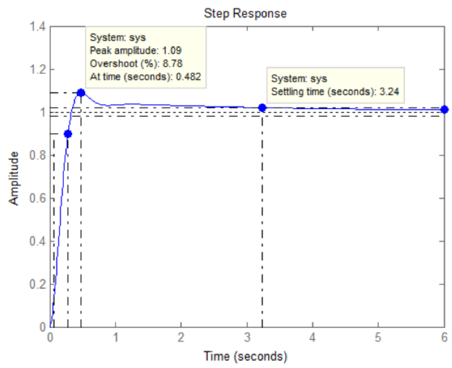
$$\begin{aligned} K_P &= 0{,}6K_U = 531 \\ K_I &= \frac{1{,}2K_U}{T_U} = 1280 \\ K_D &= \frac{0{,}6K_U T_U}{8} = 55{,}1. \end{aligned}$$

So using Ziegler-Nichols tuning the PID controller becomes

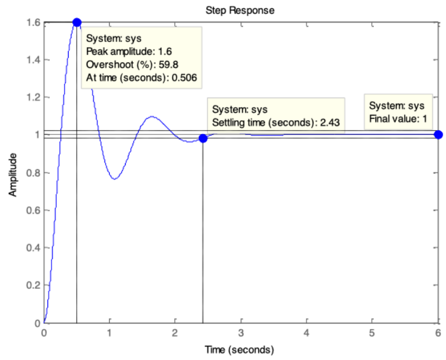
$$G_c(s) = 531 + \frac{1280}{s} + 55{,}1s = \frac{55{,}1s^2 + 531s + 1280}{s}.$$

This gives the step response on Figure 11.6b, where it is compared with the result from the manual tuning on Figure 11.6a.

On Figure 11.6 the step responses from the manually tuned on the Ziegler-Nichols tuned PID controllers are



(a) Step response from Manual tuning.



(b) Step response from Ziegler-Nichols tuning.

Figure 11.6: Comparison of step responses between manually tuned and Ziegler-Nichols tuned PID controller.

shown. Here, we see that the manual tuning has much less overshoot but a bit longer settling time than the Ziegler-Nichols tuned PID controller.

Example: Automobile Velocity Control

We define $y(t)$ to be the relative velocity between two vehicles, and our plant is modeled as

$$G(S) = \frac{1}{(s+2)(s+8)}$$

so the open-loop system has two stable poles at $s = -2$ and $s = -8$. The controller must satisfy four requirements:

1. Zero steady-state error to a step input
2. Steady-state error to a ramp input less than 25%
3. Percent overshoot less than 5%
4. Settling time less than 1,5 s

We start by considering the first design spec: zero steady state error to a step. The plant is

$$G(s) = \frac{1}{(s+2)(s+8)}$$

so we have no poles at the origins and hence a type 0 system. For zero steady state error to a step input, the open-loop system must be at least type 1, meaning it needs one integrator $1/s$. Therefore, the controller must be a type 1 controller to provide this integrator. For this, we choose a PI controller:

$$G_c(s) = K_P + \frac{K_I}{s} = \frac{K_P s + K_I}{s}.$$

For the second spec; the ramp error condition, we remember that steady-state error to a ramp input is related to the velocity error constant:

$$K_v = \lim_{s\to 0} sG_c(s)G(s).$$

We require that the error is less than 25% so $K_v > 1/0{,}25 = 4$. Now, using the PI controller and the plant:

$$G_C(s) = \frac{K_P s + K_I}{s} \qquad G(s) = \frac{1}{(s+2)(s+8)}.$$

Then:

$$K_v = \lim_{s\to 0} s\frac{K_P s + K_I}{s}\frac{1}{(s+2)(s+8)} = \lim_{s\to 0}\frac{K_P s + K_I}{(s+2)(s+8)}.$$

At $s = 0$:

$$K_v = \frac{K_I}{2\cdot 8} = \frac{K_I}{16} \implies \frac{K_I}{16} > 4 \implies K_I > 64.$$

For the overshoot condition, we look at the damping ratio ξ. The requirement here is $M_P < 5\%$, which corresponds approximately to $\xi \geq 0{,}69$. So the closed-loop dominant poles should lie in the region of the s-plane corresponding to a damping ratio of at least 0,69.
To look at the settling time condition we remember that 2% settling time is approximately:

$$T_s \approx \frac{4}{\xi\omega_n}.$$

We require $T_s \leq 1{,}5$, so

$$\frac{4}{\xi\omega_n} \leq 1{,}5$$
$$\xi\omega_n \geq 2{,}67.$$

The real part of a complex pole pair is $\Re(s) = -\xi\omega_n$, so the dominant poles should be to the left of $s \approx -2{,}6$.
The open-loop transfer function is:

$$G_c(s)G(s) = \frac{K_P s + K_I}{s(s+2)(s+8)}.$$

The closed-loop transfer function is:

$$T(S) = \frac{G_c(s)G(s)}{1 + G_c(s)G(s)}.$$

Substituting gives:

$$T(s) = \frac{K_P s + K_I}{s(s+2)(s+8) + K_P s + K_I} = \frac{K_P s + K_I}{s^3 + 10s^2 + (16 + K_P)s + K_I}.$$

We now apply the Routh-Hurwitz criterion to the characteristic equation:

$$s^3 + 10s^2 + (16 + K_P)s + K_I = 0.$$

The Routh table gives the stability conditions:

$$K_I > 0$$
$$K_P > \frac{K_I}{10} - 16.$$

So to be stable K_I must be positive and K_P must be large enough relative to K_I. From spec 2 we have $K_I > 64$, so

$$K_I > 64$$
$$K_P > \frac{K_I}{10} - 16.$$

The open-loop transfer function can be written as

$$G_c(s)G(s) = K_P\frac{s + \frac{K_I}{K_P}}{s(s+2)(s+8)}.$$

Which has poles at $s = 0$, $s = -2$, $s = -8$ and a zero at $s = -K_I/K_P$. The asymptote angles are $\phi_A = +90°, -90°$. The asymptote centroid is

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n_p - n_z} = \frac{0-2-8-\left(-\frac{K_I}{K_P}\right)}{2} = -5 + \frac{K_I}{2K_P}.$$

We now use spec 4 to choose a location for our zero. To meet the settling time requirement, the dominant poles should be left of $s = -2{,}6$. We use the asymptote centroid as a root-locus design guide and require:

$$\begin{aligned} \sigma_A &< -2{,}6 \\ -5 + \frac{K_I}{2K_P} &< 2{,}4 \\ \frac{K_I}{K_P} &< 4{,}8. \end{aligned}$$

We choose $K_I/K_P = 2{,}5$, which means the PI zero is located at $s = -2{,}5$ so the root locus becomes:

$$1 + K_P \frac{s+2{,}5}{s(s+2)(s+8)} = 8.$$

With the zero fixed at $-2{,}5$, the root locus can be plotted as shown on Figure 11.7a, where the desired region is shaded. We conclude that $K_P < 30$ is required.
Therefore our final gain conditions are

$$\begin{aligned} K_I &> 64 \\ K_P &< 30 \\ \frac{K_I}{K_P} &= 2{,}5. \end{aligned}$$

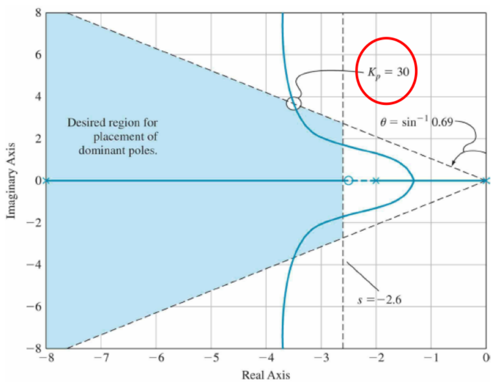
Therefore we need $2{,}5K_P > 64 \implies K_P > 25{,}6$. Together with $K_P < 30$, a reasonable choice is $K_P = 26$, and then $K_I = 2{,}5 \cdot 26 = 65$, so our final PID controller is

$$G_c(s) = 26 + \frac{65}{s} = \frac{26s+65}{s}.$$

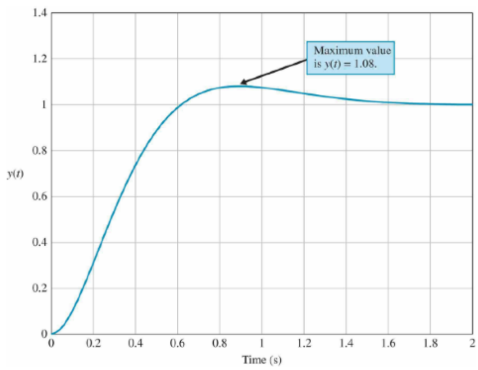
The final closed-loop transfer function is then:

$$T(s) = \frac{26s+65}{s^3+10s^2+42s+65}.$$

The final step response is shown on Figure 11.7b.



(a) Root Locus with fixed zero at $-2{,}5$.



(b) Final step response.